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On Combiners for Verifiable Random Functions

Bachelor Thesis

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"The wise man does at once what the fool does finally."

— Niccolò Machiavelli

Abstract

Verifiable random functions (VRFs) are an extension of pseudorandom functions (PRFs), one can evaluate the function using a secret key to obtain random-looking images and corresponding proofs. Given the corresponding public key one can verify whether a given image was generated from an input if given a suitable proof.

In this thesis we study ways to combine multiple instances of VRFs in a black-box fashion. Specifically we study the existence of robust black-box VRF-combiners, constructions which are given oracle access to a set of candidate schemes, of which a sufficiently large (unknown) subest implements the VRF securely, and implements a secure VRF.

While there exist simple constructions of such robust black-box combiners for the related PRF primitive even if only one candidate implements a secure PRF, we show that there exists no robust black-box VRF-combiners if not at least half of the candidate schemes are secure, even if we relax the required security properties from the combiner.

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Chapter 1

Introduction

1.1 Motivation and Previous Work

Pseudorandom Functions. A pseudorandom function (PRF) [GGM86] is a deterministic algorithm that takes as input a secret key k and a point x to produce an output y that is indistinguishable from a uniformly random bitstring of the same length. More precisely, no efficient adversary, modeled by a probabilistic polynomial-time Turing machine, can distinguish the outputs of a PRF from those of a truly random function, even after observing a polynomial number of input-output pairs. PRFs are fundamental in modern cryptography and are used to construct, among others, symmetric encryption schemes [KL14], message authentication codes [BCK96] and key derivation functions [Kra10].

Verifiable Random Functions. A verifiable function (VF) is a public-key primitive that allows the holder of a secret key sk to evaluate an input x and obtain an image-proof pair (y, π), such that any party holding the corresponding public key pk can verify that y was correctly computed from x, if given the proof π. We require two properties of the verification algorithm: *correctness*, stating that honestly generated images are accepted by the verification algorithm (always, or except with negligible probability) and *unique provability*, ensuring that for each input there exists at most one image y which is provable.

In this work, we study extensions of VFs, primarily the verifiable random function (VRF) introduced by Micali, Rabin and Vadhan [MRV99], which additionally require the images y to be *pseudorandom* to any efficient adversary without access to the corresponding proof π. Analogously to the pseudorandomness definition for PRFs, we require that no efficient adversary can distinguish the images y from images y sampled uniformly at random from the image codomain, even after observing polynomially many input-proof

pairs for other evaluation points. VRFs can be seen as a public-key analogue of PRFs with verifiability requirements.

In practice, VRFs are used to implement signatures with strong verifiability guarantees [GO93], resettable zero-knowledge proofs [MR01], lottery systems [MR02, LBM20], e-cash systems [ASM07, BCKL09], updatable zero-knowledge databases [Lis05] and transaction escrow schemes [JS04]. They play a significant role in Proof-of-Stake blockchains [KRDO17, DGKR18], specifically in distributed leader election protocols [BASV23].

Additionally, we study VFs satisfying weaker notions of pseudorandomness such as weak pseudorandomness and unpredictability, which give rise to verifiable unpredictable functions (VUFs) [MRV99] and weak verifiable random functions (WVRFs) [BGRV09] respectively. VUFs additionally require that the evaluation of the VF on a fresh evaluation point cannot be predicted by efficient adversaries except with negligible probability. WVRFs require that pseudorandomness holds, but only on uniformly random evaluation points x, instead of a point chosen by the adversary.

Robust Combiners. In general, constructions satisfying computational security properties rely on cryptographic hardness assumptions. To hedge against the breakdown of such assumptions, and the subsequent failure of the constructions built on top of them, it is desirable to base the security of a system on multiple hardness assumptions, such that security is compromised only if several assumptions fail simultaneously. Robust $(n;k)$ combiners [HKN$^+$05] are constructions that combine n candidate schemes, each implementing a given primitive, into a single scheme that remains secure whenever at least k of the candidates are secure. The notion of combiners has gained traction with the rise of post-quantum and hybrid cryptography, where the goal is to build constructions based on both emerging quantum-resistant assumptions and well-tested classical ones, ensuring that security is preserved even if one class of assumptions proves vulnerable.

A more restricted notion is given by robust $(n;k)$ black-box combiners [HKN$^+$05], which are only given oracle access to the n candidates, i.e. they may query the candidates on inputs to obtain the output but cannot inspect their underlying code. The motivation behind robust black-box combiners is to ensure that the combiner derives its security solely from evaluations of the underlying candidates, rather than by modifying them or independently implementing a secure instance of the primitive from scratch.

Robust black-box combiners have been considered for a variety of cryptographic primitives, such as oblivious transfer [HKN$^+$05], non-interactive zero-knowledge proofs [CMVX25], indistinguishability obfuscation and witness encryption [AJN$^+$16]. Additionally, folklore constructions exist for PRF, one-way functions and key agreement [HKN$^+$05], symmetric encryption and

public-key encryption. Black-box combiners for VRFs have, to the best of our knowledge, not been studied.

1.2 PRF Combiners

VRFs can be viewed as a public-key version of PRFs with an additional verifiability requirement. Simple robust black-box PRF combiners are known for PRFs. One such combiner evaluates the input x under each candidate scheme to obtain images y_1, y_2 and returns their XOR $y = y_1 \oplus y_2$. We illustrate the construction in the following schematic.

$\underline{\mathsf{Eval}_{\mathsf{PRF}}^{\mathcal{O}_1,\mathcal{O}_2}(k,x)}$

1 : $y_1 \leftarrow \mathcal{O}_1(k,x)$
2 : $y_2 \leftarrow \mathcal{O}_2(k,x)$
3 : **return** $y_1 \oplus y_2$

Figure 1.1: Robust $(2;1)$ Black-Box PRF-combiner initialized with candidate schemes $\mathcal{O}_1, \mathcal{O}_2$.

Pseudorandomness of the combined scheme follows from a standard reductionist argument: Given an adversary $\mathcal{A}$ against the combiner, we construct an adversary $\mathcal{B}$ against each individual candidate that runs in the same time as $\mathcal{A}$ and breaks the pseudorandomness of the candidate if $\mathcal{A}$ breaks the combiner's pseudorandomness. By contraposition, if at least one candidate is pseudorandom then the combiner is pseudorandom.

The reduction against candidate scheme $\mathcal{O}_i$ has access to the pseudorandomness oracle $\mathcal{O}_{\text{eval}}$ of the pseudorandomness security experiment against $\mathcal{O}_i$ and internally simulates the other candidate $\mathcal{O}_{2-i}$, answering queries from $\mathcal{A}$ with $\mathcal{O}_{\text{eval}}(x) \oplus \mathcal{O}_{2-i}(k,x)$. Note that $\mathcal{B}$ makes exactly as many queries to $\mathcal{O}_{\text{eval}}$ as $\mathcal{A}$ does during its simulation, so $\mathcal{B}$ is efficient whenever $\mathcal{A}$ is. The adversary $\mathcal{A}$ against the combined scheme outputs a guess b' for the hidden bit b, sampled by the challenger uniformly at random, attempting to distinguish the real world from the random world. The reduction simply forwards this guess to its own challenger. If $b = 0$, the adversary $\mathcal{A}$ is simulated perfectly, as it receives images, computed exactly as they would be in the pseudorandomness security experiment against the combiner. If $b = 1$ then $\mathcal{O}_{\text{eval}}(x)$ returns images sampled u.a.r. from the set of bitstrings with appropriate length. Since $\mathcal{O}_{\text{eval}}(x)$ is uniformly distributed, $\mathcal{O}_{\text{eval}}(x) \oplus \mathcal{O}_{2-i}(k,x)$ is uniformly distributed as well. Since $\mathcal{A}$ is perfectly simulated in both worlds, $\mathcal{B}$ wins the pseudorandomness experiment with the same advantage as $\mathcal{A}$.

The existence of a robust $(2;1)$ black-box PRF-combiner naturally leads to a robust $(n;1)$ black-box PRF combiner for any $n \in \mathbb{N}^+$, by iteratively

applying the pairwise construction with a fresh candidate at each step. Since there exist simple robust $(n;1)$ black-box PRF-combiners, one might hope that an analogous approach extends to VRFs. This turns out not to be the case as we show in Theorem 3.5 that there exists no robust $(2;1)$ black-box VRF-combiner.

1.3 Our Contribution

Robust Black-Box DAG Combiner. We introduce the notion of robust black-box DAG combiners, formally defined in Definition 3.1. A robust black-box DAG combiner is a black-box combiner whose implemented algorithms are each specified by a directed acyclic graph (DAG) $\mathcal{G}$. Execution of the algorithm proceeds along the DAG starting from a designated source vertex s and ending in a designated sink vertex t. At each intermediate vertex $v \in V(\mathcal{G})$, a probabilistic polynomial-time Turing machine, without access to any oracles, computes a query input $\mathbf{q}_v$ from the outputs of all preceding vertices, which is subsequently queried to an oracle algorithm of a candidate specified by a vertex labeling $\ell(v)$ to obtain an output $\mathbf{x}_v$. DAG combiners make the structure of the black-box combiner more explicit by decomposing it into interleaved oracle queries to the candidates and intermediate (oracle independent) computation steps. We illustrate the concept of a robust black-box DAG combiner in Figure 1.2 using the robust black-box PRF combiner from Section 1.2 as an example.

We relate robust black-box DAG combiners with general robust black-box combiners by showing that the two notions are equivalent in the sense that for every general robust black-box combiner there exists a robust black-box DAG combiner whose algorithms behave identically (the converse holds definitionally). We state this result formally in Theorem 3.2. The robust black-box DAG combiner framework provides a more structured representation of robust black-box combiners which simplifies our proof of Theorem 3.5.

Robust Black-Box VRF-Combiners. With the robust black-box DAG combiner framework in place, we consider the simplest robust black-box VRF-combiner. Namely, the robust $(2;1)$ black-box VRF-combiner, where only two candidates are combined and security of the combined scheme is required to hold, if at least one of the candidates is secure. We prove that no such combiner exists in Theorem 3.5. As we can use a robust $(n;\lfloor n/2 \rfloor)$ black-box VRF-combiner to construct a robust $(2;1)$ black-box combiner, we extend the impossibility of a robust $(2;1)$ black-box VRF-combiner in Theorem 3.5 to the impossibility of an infinite family of robust $(n;\lfloor n/2 \rfloor)$ black-box combiners for all $n \in \mathbb{N}$ in Lemma 3.20.

Finally, we investigate robust black-box combiners under strengthened as-

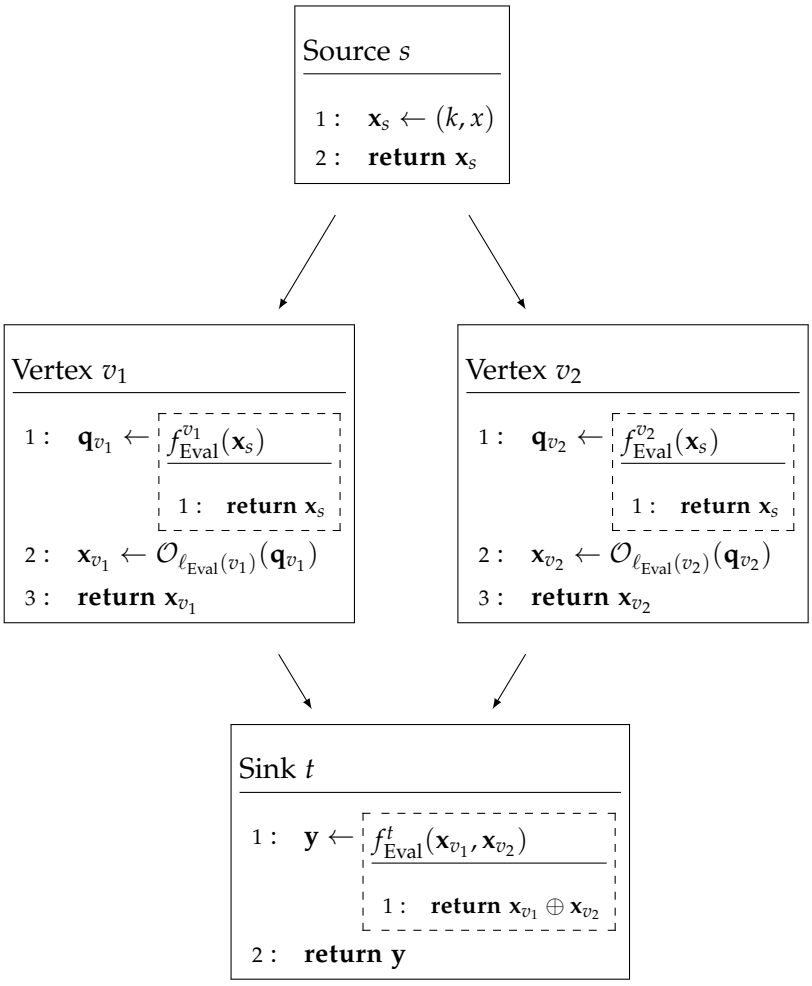


Figure 1.2: Robust $(2;1)$ Black-Box DAG PRF-combiner. For every vertex v, we first combine the outputs of all predecessors using a ppt. Turing machine f^v_{Eval}. The resulting value $\mathbf{q}_v$ is then queried to the corresponding oracle algorithm $\mathcal{O}_{\ell(v)}$. The auxiliary source vertex simply outputs the input $\mathbf{x}_s$, while the sink vertex t doesn't query any oracle but only computes the final output $\mathbf{y}$. In the formal definition, each intermediate Turing machine f^v_{Eval} takes as input a set of labeled tuples $(v, \mathbf{x}_v)$, pairing each predecessor vertex with its corresponding output. For the sake of readability, we simply list the outputs as arguments.

sumptions on the candidate schemes or relaxed requirements on the combiner construction. We show that if all candidate schemes are assumed to be uniquely provable and perfectly correct, then a simple robust $(n;1)$ black-box VRF-combiner exists. This construction extends the robust $(n;1)$ black-box PRF-combiner presented in Section 1.2.

Chapter 2

Notation and Preliminaries

2.1 Notation

For a positive integer n define $[n] = \{1, \ldots, n\}$. We use $\lambda \in \mathbb{N}$ exclusively to denote the security parameter. We use $\mathsf{poly}(\lambda)$ to denote the set of functions that are polynomial in λ in the sense of Definition 2.1 and $\mathsf{negl}(\lambda)$ to denote the set of functions that are negligible in λ in the sense of Definition 2.2. We call a function f *overwhelming* if $1 - f(\lambda) \in \mathsf{negl}(\lambda)$. We use $\leftarrow$ for simple assignment. For a set S, we write $s \leftarrow_\$ S$ to indicate that s is sampled uniformly at random (u.a.r.) from set S. For a probabilistic algorithm $\mathcal{A}$, we write $y \leftarrow_\$ \mathcal{A}(x)$ to denote assigning to y the output of $\mathcal{A}$ on input x. We use ppt. to abbreviate *probabilistic polynomial time* in the context of Turing machines. Finally, we use bold letters (e.g. $\mathbf{x}$) to indicate that a symbol represents multiple values, such as tuples or vectors.

Definition 2.1 (Polynomial Functions) *A function $p : \mathbb{N} \to \mathbb{R}$ is polynomial if*

$$\exists c, d \in \mathbb{N}, \exists n_0 \in \mathbb{N}, \forall n > n_0, p(n) \leq c \cdot n^d$$

Definition 2.2 (Negligible Functions) *A function $\varepsilon : \mathbb{N} \to \mathbb{R}$ is negligible if*

$$\forall p \in \mathsf{poly}(\lambda), \exists n_0 \in \mathbb{N}, \forall n > n_0, |\varepsilon(n)| < \frac{1}{p(n)}$$

2.2 Verifiable Functions

A verifiable function (VF) VF is a cryptographic primitive that implements algorithms $\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}$ and $\mathsf{Vfy}_{\mathsf{VF}}$. A user can generate a key pair $(\mathsf{sk}, \mathsf{pk})$ using $\mathsf{Gen}_{\mathsf{VF}}$, consisting of a secret key sk and a public key pk. Subsequently, the user can use the secret key sk to evaluate arbitrary points $x \in \{0,1\}^\lambda$ using $\mathsf{Eval}_{\mathsf{VF}}$ and use the public key pk to verify a triple (x, y, π) using $\mathsf{Vfy}_{\mathsf{VF}}$. We give a formal definition in Definition 2.3. The definition is heavily inspired by the definition of a VRF in [HJ16]. To reduce redundancy, we define additional security properties such as pseudorandomness and unpredictability separately.

Definition 2.3 (Verifiable Function) *A verifiable function (VF)* VF *is a triple of efficient (ppt.) algorithms* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *with the syntax:*

- $\mathsf{Gen}_{\mathsf{VF}}$*: The key generation algorithm takes as input the unary encoding of the security parameter and produces a pair* $(\mathsf{sk}, \mathsf{pk}) \in \mathcal{SK} \times \mathcal{PK}$ *consisting of a secret key and a public key.*
- $\mathsf{Eval}_{\mathsf{VF}}$*: The evaluation algorithm takes as input a secret key* $\mathsf{sk} \in \mathcal{SK}$ *and an evaluation point* $x \in \{0,1\}^\lambda$ *and outputs a pair* (y, π) *consisting of an image* $y \in \mathcal{Y}$ *and a proof* $\pi \in \mathcal{P}$*.*
- $\mathsf{Vfy}_{\mathsf{VF}}$*: The verification algorithm takes as input a public key* $\mathsf{pk} \in \mathcal{PK}$*, an evaluation point* $x \in \{0,1\}^\lambda$*, an image* $y \in \mathcal{Y}$ *and a proof* $\pi \in \mathcal{P}$ *and returns a bit* b *indicating acceptance or rejection.*

We use the notation $\mathsf{Eval}^y_{\mathsf{VF}}$ and $\mathsf{Eval}^\pi_{\mathsf{VF}}$ to denote the algorithm $\mathsf{Eval}_{\mathsf{VF}}$ projected on the respective output component. Note that a VF is a purely syntactic concept. We define various semantic properties for a VF to arrive at specific cryptographic primitives, e.g. VRFs. First, we define ε-correctness, a property that guarantees that over random key pairs and evaluation points, the verification algorithm accepts a correctly computed image-proof pair with probability at least ε.

Definition 2.4 (ε-Correctness) *A verifiable function* $\mathsf{VF} = (\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is* ε*-correct if for all security parameters* $\lambda \in \mathbb{N}$ *we have*

$$\Pr\left[\begin{array}{l}(\mathsf{sk}, \mathsf{pk}) \leftarrow_\$ \mathsf{Gen}_{\mathsf{VF}}(1^\lambda) \\ \qquad x \leftarrow_\$ \{0,1\}^\lambda \qquad : \mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y, \pi) = 1 \\ (y, \pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)\end{array}\right] \geq \varepsilon(\lambda)$$

We say a VF VF *is perfectly correct, or simply correct, if it is* 1*-correct, and statistically correct if* ε *is an overwhelming function.*

Next, we define unique provability. A VF is uniquely provable if no more than one image can be proven for any input.

Definition 2.5 (Unique Provability, [Bra25]) *A VF* $\mathsf{VF} = (\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is uniquely provable if for all evaluation points* $x \in \{0,1\}^\lambda$*, all public-keys* $\mathsf{pk} \in \mathcal{PK}$ *and any two image-proof pairs* $(y_0, \pi_0), (y_1, \pi_1) \in \mathcal{Y} \times \mathcal{PK}$*, we have*

$$\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y_0, \pi_0) = 1 \land \mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y_1, \pi_1) = 1 \implies y_0 = y_1$$

We define pseudorandomness of a VF, a computational property stating that no efficient adversary $\mathcal{A}$ can distinguish an image y corresponding to the evaluation of an input x^*, chosen by $\mathcal{A}$, from a uniformly random element $y \leftarrow_\$ \mathcal{Y}$. This must hold even if the adversary is given evaluations at polynomially many points $x \neq x^*$. The definition is inspired by [Bra25].

Definition 2.6 ($(\mathsf{t}, \varepsilon)$-Pseudorandomness) *A VF* $\mathsf{VF} = (\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is* $(\mathsf{t}, \varepsilon)$*-pseudorandom if for all security parameters* $\lambda \in \mathbb{N}$ *and all adversaries* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ *running in time at most* $\mathsf{t}(\lambda)$*, the advantage of* $\mathcal{A}$ *in the pseudorandomness security experiment in Figure 2.1 is at most* $\varepsilon(\lambda)$*, i.e.*

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a verifiable function VF *is pseudorandom if it is* $(\mathsf{t}, \varepsilon)$*-pseudorandom with* $\varepsilon \in \mathsf{negl}(\lambda)$ *and* $\mathsf{t} \in \mathsf{poly}(\lambda)$*.*

Next, we define two weaker notions of pseudorandomness. While (strong) pseudorandomness requires the evaluation at adversarially chosen points to be indistinguishable from uniformly random elements of $\mathcal{Y}$, weak pseudorandomness relaxes this by requiring indistinguishability only for uniformly random challenges $x^* \leftarrow_\$ \{0,1\}^\lambda$.

Definition 2.7 ($(\mathsf{t}, \varepsilon)$-Weak Pseudorandomness) *A VF* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is* $(\mathsf{t}, \varepsilon)$*-weak pseudorandom if for all security parameters* $\lambda \in \mathbb{N}$ *and all adversaries* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ *running in time at most* $\mathsf{t}(\lambda)$*, the advantage of* $\mathcal{A}$ *in the weak pseudorandomness security experiment in Figure 2.1 is at most* $\varepsilon(\lambda)$*, i.e.*

$$\mathbf{Adv}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a verifiable function VF *is weak pseudorandom if it is* $(\mathsf{t}, \varepsilon)$*-weak pseudorandom with* $\varepsilon \in \mathsf{negl}(\lambda)$ *and* $\mathsf{t} \in \mathsf{poly}(\lambda)$*.*

Alternatively, a VF can be unpredictable. A VF is unpredictable if no adversary can predict the image y of the evaluation at a challenge x^* chosen by the adversary. This must hold even if the adversary sees evaluations at polynomially many points $x \neq x^*$. Note that if an adversary can predict

the output at x^*, it can also distinguish the output from a random element $y \leftarrow_\$ \mathcal{Y}$ with non-negligible probability, making unpredictability a weaker property than pseudorandomness.

Definition 2.8 ((t,ε)-Unpredictability) *A VF* $\mathsf{VF} = (\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is* (t,ε)*-unpredictable if for all security parameters* $\lambda \in \mathbb{N}$ *and all adversaries* $\mathcal{A}$ *running in time at most* $t(\lambda)$*, the advantage of* $\mathcal{A}$ *in the unpredictability security experiment in Figure 2.1 is at most* $\varepsilon(\lambda)$*, i.e.*

$$\mathbf{Adv}^{\mathrm{up}}_{\mathsf{VF}}(\lambda,\mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{up}}_{\mathsf{VF}}(\lambda,\mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a verifiable function VF *is unpredictable if it is* (t,ε)*-unpredictable with* $\varepsilon \in \mathsf{negl}(\lambda)$ *and* $t \in \mathsf{poly}(\lambda)$*.*

Experiment $\mathbf{Exp}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda,\mathcal{A})$

1 : $b \leftarrow_\$ \{0,1\}$
2 : $(\mathsf{sk},\mathsf{pk}) \leftarrow_\$ \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
3 : $S \leftarrow \varnothing$
4 : $(x^*,\sigma) \leftarrow_\$ \mathcal{A}_1^{\mathcal{O}_{\mathrm{eval}}}(1^\lambda,\mathsf{pk})$
5 : $(y_0,\pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk},x^*)$
6 : $y_1 \leftarrow_\$ \mathcal{Y}$
7 : $b' \leftarrow_\$ \mathcal{A}_2(1^\lambda,x^*,\sigma,y_b)$
8 : **return** $b = b' \wedge x^* \notin S$

Experiment $\mathbf{Exp}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda,\mathcal{A})$

1 : $b \leftarrow_\$ \{0,1\}$
2 : $(\mathsf{sk},\mathsf{pk}) \leftarrow_\$ \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
3 : $S \leftarrow \varnothing$
4 : $\sigma \leftarrow_\$ \mathcal{A}_1^{\mathcal{O}_{\mathrm{eval}}}(1^\lambda,\mathsf{pk})$
5 : $x^* \leftarrow_\$ \{0,1\}^\lambda$
6 : $(y_0,\pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk},x^*)$
7 : $y_1 \leftarrow_\$ \mathcal{Y}$
8 : $b' \leftarrow_\$ \mathcal{A}_2^{\mathcal{O}_{\mathrm{eval}}}(1^\lambda,x^*,\sigma,y_b)$
9 : **return** $b = b' \wedge x^* \notin S$

Experiment $\mathbf{Exp}^{\mathrm{up}}_{\mathsf{VF}}(\lambda,\mathcal{A})$

1 : $(\mathsf{sk},\mathsf{pk}) \leftarrow_\$ \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
2 : $S \leftarrow \varnothing$
3 : $(x^*,y^*) \leftarrow_\$ \mathcal{A}^{\mathcal{O}_{\mathrm{eval}}}(1^\lambda,\mathsf{pk})$
4 : $(y,\pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk},x^*)$
5 : **return** $y^* = y \wedge x^* \notin S$

Oracle $\mathcal{O}_{\mathrm{eval}}(x)$

1 : $S \leftarrow S \cup \{x\}$
2 : $(y,\pi) \leftarrow_\$ \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk},x)$
3 : **return** y

Figure 2.1: The pseudorandomness security experiment $\mathbf{Exp}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda,\mathcal{A})$ from Definition 2.6, the weak pseudorandomness security experiment $\mathbf{Exp}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda,\mathcal{A})$ from Definition 2.7 and the unpredictability security experiment $\mathbf{Exp}^{\mathrm{up}}_{\mathsf{VF}}(\lambda,\mathcal{A})$ from Definition 2.8. The oracle $\mathcal{O}_{\mathsf{eval}}$ allows the adversary to evaluate arbitrary points $x \in \{0,1\}^\lambda$ under the hidden secret key sk. Differences between the pseudorandomness security experiment and the weak pseudorandomness security experiment are highlighted in grey.

We now define verifiable random functions (VRFs), weak verifiable random functions (WVRFs) and verifiable unpredictable functions (VUFs). These are VFs that are correct, uniquely provable and pseudorandom, weak pseudorandom or unpredictable respectively.

Definition 2.9 (Verifiable Random Functions) *A verifiable random function* $\mathsf{VRF} = (\mathsf{Gen}_{\mathsf{VRF}}, \mathsf{Eval}_{\mathsf{VRF}}, \mathsf{Vfy}_{\mathsf{VRF}})$ *is a VF that satisfies perfect correctness (Definition 2.4), unique provability (Definition 2.5) and pseudorandomness (Definition 2.6).*

Definition 2.10 (Weak Verifiable Random Functions) *A weak verifiable random function* $\mathsf{WVRF} = (\mathsf{Gen}_{\mathsf{WVRF}}, \mathsf{Eval}_{\mathsf{WVRF}}, \mathsf{Vfy}_{\mathsf{WVRF}})$ *is a VF that satisfies perfect correctness (Definition 2.4), unique provability (Definition 2.5) and weak pseudorandomness (Definition 2.7).*

Definition 2.11 (Verifiable Unpredictable Functions) *A verifiable unpredictable function* $\mathsf{VUF} = (\mathsf{Gen}_{\mathsf{VUF}}, \mathsf{Eval}_{\mathsf{VUF}}, \mathsf{Vfy}_{\mathsf{VUF}})$ *is a VF that satisfies perfect correctness (Definition 2.4), unique provability (Definition 2.5) and unpredictability (Definition 2.8).*

2.3 Robust Combiners

For two natural numbers n, k we define the robust $(n;k)$ black-box combiner. A robust $(n;k)$ black-box combiner for a primitive P is a construction Π_{P}, modeled by a ppt. oracle Turing machine, that is given access to n oracles, the candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$. If at least k candidate schemes securely implement the primitive P, then the combiner $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ also implements a secure instance of P.

The intuition behind a robust black-box combiner is that the construction extracts security properties from the subset of secure candidates, even if some candidate schemes are insecure. However, note the following issue with this definition: If there exists a secure instance of P, then there always exists a robust $(n;0)$ black-box P-combiner that simply ignores its candidates and implements P directly. Note that such a combiner trivially meets all requirements of the definition. Proving that there exists no robust black-box combiner would therefore require showing that no secure instance of P exists.

To circumvent this problem, we explicitly force the combiner to derive its security from the subset of secure candidates. We achieve this, at least for computational security notions, by requiring the security properties to hold relative to a PSPACE-complete oracle. This means that an adversary against the combiner can efficiently decide membership in a PSPACE-complete language, enabling the adversary to break computational hardness assumptions while still being limited to polynomially many oracle queries. We note that this notion of a combiner does not resolve the issue when all security properties of the cryptographic primitive are information-theoretic.

Definition 2.12 (Robust $(n;k)$ Black-Box P-Combiner) *A robust $(n;k)$ black-box combiner for primitive P with k algorithms is a tuple of ppt. oracle Turing machines $\Pi_{\mathsf{P}} = (\mathsf{Alg}_{\mathsf{P}}^1, \ldots, \mathsf{Alg}_{\mathsf{P}}^k)$ such that the following holds: If there exists a ppt. adversary $\mathcal{A}^{\mathcal{O}_1,\ldots,\mathcal{O}_n,\mathsf{PSPACE}}$ that breaks the security of $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$, then there exist $n-k+1$ ppt. adversaries $\mathcal{B}_1^{\mathcal{O}_{\ell(1)}}, \ldots, \mathcal{B}_{n-k+1}^{\mathcal{O}_{\ell(n-k+1)}}$ and an injective function $\ell : [n-k+1] \to [n]$ such that for any $i \in [n-k+1]$ the adversary $\mathcal{B}_i^{\mathcal{O}_{\ell(i)}}$ breaks the security of the candidate $\mathcal{O}_{\ell(i)}$.*

Remark. If primitive P has composite security properties, as it is the case for VRFs, we say that an adversary breaks the security of P if it breaks any of the required security properties. For computational properties such as pseudorandomness, this means winning the corresponding security experiment with non-negligible advantage. For information-theoretic or statistical properties such as unique provability or correctness, this means the property does not hold.

Chapter 3

Results

3.1 Robust Black-Box DAG Combiner

Intuitively, a robust black-box combiner for a primitive P that implements algorithms $(\mathsf{Alg}_{\mathsf{P}}^1, \ldots, \mathsf{Alg}_{\mathsf{P}}^p)$ operates on n candidate schemes by adaptively interleaving calls to the candidate oracles with ppt. processing, combining the candidate outputs in a way that preserves the security properties of the secure subset of oracles, provided there are sufficiently many. For certain arguments, it is helpful to model the combiner's structure more explicitly. We do so by representing each algorithm implemented by the combiner as a directed acyclic graph (DAG) $\mathcal{G}$ that reflects the computation.

In addition to the DAG, we specify dedicated source and sink vertices s and t, a vertex labeling $\ell : V(\mathcal{G}) \setminus \{s,t\} \to [n] \times [|\mathsf{P}|]$ that maps every non-source and non-sink vertex to a tuple (i,j) indicating which candidate is queried on which oracle, and a ppt. Turing machine f^v for every vertex $v \in V(\mathcal{G}) \setminus \{s,t\}$ that computes the query input to the candidate and oracle $\ell(v)$ given the oracle responses of the predecessors $\mathcal{N}_{\mathcal{G}}^-(v)$. The source vertex s corresponds to the input $\mathbf{x}$ to the algorithm, while the sink vertex t is associated with a ppt. Turing machine f^t that computes the final output $\mathbf{y}$ of the combiner.

A robust black-box DAG combiner algorithm is executed by tracing the computation through the corresponding DAG $\mathcal{G}$. The input to each oracle query is computed from the responses of the oracles at preceding vertices using the Turing machine f^v associated with the respective vertex $v \in V(\mathcal{G})$. The output of the algorithm is given by the output of f^t on the collected inputs. We assume that the random tapes of the intermediate Turing machines are initialized identically to the random tape of the combiner Turing machine.

A robust black-box DAG combiner has to satisfy the requirements that the algorithm specified by the DAG description is a robust black-box combiner. We formally define robust black-box DAG combiners below.

Definition 3.1 (Robust $(n;k)$ Black-Box DAG P-Combiner) *A robust $(n;k)$ black-box DAG combiner for a primitive P is a robust $(n;k)$ black-box combiner Π_{P} in the sense of Definition 2.12, i.e., a tuple of algorithms $(\mathsf{Alg}_{\mathsf{P}}^1, \ldots, \mathsf{Alg}_{\mathsf{P}}^p)$ where every algorithm $\mathsf{Alg}_{\mathsf{P}}$ is specified by a tuple of the following form*

$$\left(\mathcal{G}, s, t, \ell, \{f^v\}_{v \in V(\mathcal{G}) \setminus \{s\}}\right)$$

Here, $\mathcal{G}$ is a DAG, s and t are vertices in $V(\mathcal{G})$ such that s has no incoming edges and t has no outgoing edges, $\ell : V(\mathcal{G}) \to [n] \times [|\mathsf{P}|]$ is a vertex labeling and $\{f^v\}_{v \in V(\mathcal{G}) \setminus \{s\}}$ are polynomial time Turing machines. Given this tuple, the algorithm $\mathsf{Alg}_{\mathsf{P}}$ is defined as shown in Figure 3.1.

Algorithm $\mathsf{Alg}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}(\mathbf{x})$

1 : $v_1, \ldots, v_m \leftarrow$ topological ordering of $V(\mathcal{G}) \setminus \{s, t\}$
2 : $\mathbf{x}_s \leftarrow \mathbf{x}$
3 : **for** $i = 1$ to m **do**
4 : // Compute the query $\mathbf{q}_{v_i}$ to the oracle
5 : $\mathbf{q}_{v_i} \leftarrow f^{v_i}(\{(w, \mathbf{x}_w)\}_{w \in \mathcal{N}_{\mathcal{G}}^-(v_i)})$
6 : // Evaluate the oracle on $\mathbf{q}_{v_i}$
7 : $\mathbf{x}_{v_i} \leftarrow \mathcal{O}_{\ell(v_i)}(\mathbf{q}_{v_i})$
8 : **return** $f^t(\{w, \mathbf{x}_w\}_{w \in \mathcal{N}_{\mathcal{G}}^-(t)})$

Figure 3.1: Definition of the robust black-box DAG combiner algorithm $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}}$ instantiated with candidates $\mathcal{O}_1, \ldots, \mathcal{O}_n$. We use the notation $\mathcal{O}_{\ell(v)}$ for $\ell(v) = (i, j)$ to denote the oracle algorithm j of the i-th candidate $\mathcal{O}_i$, i.e., $\mathcal{O}_i.\mathsf{Alg}_{\mathsf{P}}^j$.

By definition, every robust black-box DAG combiner is a robust black-box combiner. We also show that for every robust black-box combiner there exists a canonical robust black-box DAG combiner that is *computationally equivalent*, in the sense that the implemented algorithms induce identical output distributions. We state this result formally in the following theorem, which allows us to conduct our impossibility proof for Theorem 3.5 over robust black-box DAG combiners rather than robust black-box combiners.

Theorem 3.2 (Robust $(n;k)$ Black-Box DAG Combiners) *For every robust $(n;k)$ black-box combiner (Definition 2.12) there exists a computationally equivalent robust $(n;k)$ black-box DAG combiner (Definition 3.1).*

Intuition. The idea behind the proof is to take an arbitrary robust $(n;k)$ black-box combiner Π_{P} and unroll its computation into a graph consisting of $Q(n, \lambda)$ layers, where each vertex on layer k is connected to all vertices of the preceding layers $i < k$. Each layer k consists of $n|\mathsf{P}|$ vertices corresponding to

the different oracles that can be queried as the k-th query. At every layer k, for every vertex, the combiner simulates the Turing machine of Π_{P} until the k-th oracle query. Since the Turing machine f^v cannot make any oracle queries itself, it relies on the preceding connected vertices to obtain the answer to those queries. For the i-th query, the Turing machine simply uses the output of the correct vertex on layer i to simulate the oracle query. Once we reach the k-th query and we would make a query to oracle $\mathcal{O}_{(i,j)}$ then we distinguish two cases depending on whether $\ell(v) = (i, j)$.

If the current vertex satisfies $\ell(v) = (i, j)$, we return the query $\mathbf{q}_v$, resulting in the correct query being made. Otherwise, if $\ell(v) \neq (i, j)$, we return an arbitrary query that meets the expected input format of $\mathcal{O}_{\ell(v)}$. An important detail is that we must fix the random tape ρ of the Turing machines f^v for all $v \in V(\mathcal{G}) \setminus \{s\}$, since otherwise the executions may diverge when simulating oracle queries at later points. To address this, we can simply treat the Turing machines f^v as deterministic with an additional input ρ for the randomness. For notational simplicity, we leave this implicit in what follows.

Proof (Theorem 3.2) Let Π_{P} be a fixed robust $(n;k)$ black-box combiner for primitive P, and let $\mathsf{Alg}_{\mathsf{P}}$ be an arbitrary algorithm implemented by Π_{P}. We label the candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$. We construct a robust $(n;k)$ black-box DAG combiner $\widetilde{\Pi}_{\mathsf{P}}$ that is computationally equivalent to Π_{P}.

Since Π_{P} is a robust $(n;k)$ black-box combiner, it is a ppt. Turing machine and the number of oracle queries made by Π_{P} is bounded from above by $Q \in \mathsf{poly}(n, \lambda)$. We construct the DAG $\mathcal{G}$ for the robust black-box DAG combiner description of algorithm $\mathsf{Alg}_{\mathsf{P}}$ as follows:

$$V(\mathcal{G}) = \{s, t\} \cup [Q(n, \lambda)] \times [n] \times [|\mathsf{P}|] \tag{3.1}$$

$$E(\mathcal{G}) = \big\{ \big((d_1, i_1, j_1), (d_2, i_2, j_2)\big) \in [Q(n, \lambda)] \times [n] \times [|\mathsf{P}|] \big\} \tag{3.2}$$

$$\cup \{s\} \times [Q(n, \lambda)] \times [n] \times [|\mathsf{P}|] \tag{3.3}$$

$$\cup [Q(n, \lambda)] \times [n] \times [|\mathsf{P}|] \times \{t\} \tag{3.4}$$

We define the vertex labeling $\ell(d, i, j) = (i, j)$ and the ppt. Turing machines f^v as follows. On input $\{(w, \mathbf{x}_w)_{w \in \mathcal{N}_{\mathcal{G}}^-(v)}\}$, simulate the Turing machine $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ on the input $\mathbf{x}$. For every q-th query made to the oracle (i, j), look up the values of $\mathbf{x}_{(q,i,j)}$ and continue execution with that oracle response. For the k-th query, if the query is to oracle $\ell(v)$, return the query. Otherwise, if the k-th query is to an oracle $(i, j) \neq \ell(v)$ return an arbitrary element in the domain of oracle $\ell(v)$ (this query will not be considered further in the execution). For f^t, simulate the entire execution of $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ analogously, but since no further queries are made, simply return the final result.

To show that $\widetilde{\Pi}_{\mathsf{P}}$ is a robust $(n;k)$ black-box DAG combiner, we need to show that the algorithm $\widetilde{\Pi}_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ specified by $(\mathcal{G}, s, t, \ell, \{f^v\}_{v \in V(\mathcal{G})})$ is computation-

ally equivalent to $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ and runs in time polynomial in λ, n. We first show that the algorithm is computationally equivalent.

Claim 3.3 *Let Π_{P} be a robust $(n;k)$ black-box combiner for primitive P (Definition 2.12) and let $\widetilde{\Pi}_{\mathsf{P}}$ be the combiner obtained by the procedure above. Then for all candidates $\mathcal{O}_1, \ldots, \mathcal{O}_n$ and every algorithm $\mathsf{Alg}_{\mathsf{P}}$ implemented by Π_{P} we have*

$$\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}} \equiv \widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}}$$

Where $\equiv$ denotes equality of the output distributions.

Proof (Claim 3.3) Fix a robust $(n;k)$ black-box P-combiner Π_{P}, candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$ and an algorithm $\mathsf{Alg}_{\mathsf{P}}$ implemented by Π_{P}. Let $\widetilde{\Pi}_{\mathsf{P}}$ be the robust $(n;k)$ black-box DAG P-combiner constructed as specified above, and let $\mathbf{x}$ be an arbitrary input to the algorithm $\mathsf{Alg}_{\mathsf{P}}$. We show that

$$\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}}(\mathbf{x}) \equiv \widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}}(\mathbf{x})$$

We proceed by coupling the randomness of the execution of both algorithms on $\mathbf{x}$. Specifically, we use the same random tape for the combiner's internal computation in both Π_{P} and $\widetilde{\Pi}_{\mathsf{P}}$ and for each query index d and oracle (i, j) we use the same random tape for every oracle response. Under this coupling, we show by induction on the number of queries that the outputs are identical.

We say that a vertex v on layer $d \in [Q(n, \lambda)]$ is *correct* if in the execution of $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Alg}_{\mathsf{P}}(\mathbf{x})$ the d-th oracle query is made to the oracle $\ell(v)$. We say that it is *incorrect* otherwise. Formally, we show that for every $d \in [Q(n, \lambda)]$, the output of the d-th query of $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ is equal to the output at the correct vertex on layer d of $\widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$. We proceed by strong induction over d:

Base case: Let $d = 1$. The robust black-box DAG combiner $\widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ simulates $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ until it reaches the first oracle query. Subsequently, it makes the same query $\mathbf{q}_v$ to the same oracle $\mathcal{O}_{\ell(v)}$ at the correct vertex v. Under the coupled randomness, the oracle responses are identical.

Inductive step: Assume that for all $d' < d$ the output of the correct vertex on layer d' is identical to the output of the d'-th oracle query in $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$. Let v be the correct vertex in layer d. The Turing machine f^v simulates $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ using the outputs of correct vertices on preceding layers to answer oracle queries. By the induction hypothesis, these outputs match those of $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ so f^v computes the same query input $\mathbf{q}_v$. Since the vertex is correct, it queries the same oracle on the same input under the coupled randomness. Therefore, the responses are identical. □

Claim 3.4 *Let Π_{P} be a robust $(n;k)$ black-box combiner for a primitive P (Definition 2.12) and let $\widetilde{\Pi}_{\mathsf{P}}$ be the combiner obtained by the procedure above. Then for all candidates $\mathcal{O}_1, \ldots, \mathcal{O}_n$, every algorithm $\mathsf{Alg}_{\mathsf{P}}$ implemented by $\widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ runs in time polynomial in n, λ and the size of the input to $\mathsf{Alg}_{\mathsf{P}}$.*

Proof (Claim 3.4) By construction, the DAG $\mathcal{G}$ has at most $2 + n \cdot |\mathsf{P}| \cdot Q(n, \lambda)$ vertices. Here, $Q(n, \lambda)$ is a polynomial upper bound on the number of queries of $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$, where we assume that P is constant. For every vertex $v \in V(\mathcal{G})$, the Turing machine f^v simulates $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$. In this simulation every oracle query except the last one is answered with a constant-time lookup of the query response in the predecessor outputs. The last query is made to a candidate oracle, also in constant-time. Therefore, the computation done at each vertex is bounded by $T(n, \lambda)$, where T is a polynomial bounding the runtime of $\Pi_{\mathsf{P}}.\mathsf{Alg}_{\mathsf{P}}$ from above. The overall running time is bounded by the following expression.

$$\big(2 + n \cdot |\mathsf{P}| \cdot Q(n, \lambda)\big) \cdot T(\lambda, n) \in \mathsf{poly}(\lambda, n) \qquad \square$$

It remains to show that $\widetilde{\Pi}_{\mathsf{P}}$ is a robust $(n; k)$ black-box combiner. By Claim 3.3, the algorithms implemented by $\widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ and $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ have equivalent output distributions. In particular, any adversary $\mathcal{A}^{\mathcal{O}_1,\ldots,\mathcal{O}_n,\mathsf{PSPACE}}$ against $\widetilde{\Pi}_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ is also an adversary against $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ with the same advantage. Since Π_{P} is a robust black-box combiner, by Definition 2.12, there exist an injective function $\ell : [n - k + 1] \to [n]$ and ppt. adversaries

$$\mathcal{B}_1^{\mathcal{O}_{\ell(1)}}, \ldots, \mathcal{B}_{n-k+1}^{\mathcal{O}_{\ell(n-k+1)}}$$

such that for any $i \in [n - k + 1]$ the adversary $\mathcal{B}_i^{\mathcal{O}_{\ell(i)}}$ breaks the security of the candidate $\mathcal{O}_{\ell(i)}$. Hence $\widetilde{\Pi}_{\mathsf{P}}$ is a robust $(n; k)$ black-box combiner. $\square$

Example. We illustrate the concept of a robust $(n; k)$ black-box DAG combiner in Figure 1.2 using the example of the robust $(2; 1)$ black-box PRF-combiner in Section 1.2. Note that the robust black-box DAG combiner shown in Figure 1.2 is not the same as one would have obtained by applying the procedure from the proof of Theorem 3.2.

3.2 Impossibility of Robust Black-Box VRF-Combiner

We consider the possibility of combining VRFs in a black-box fashion. Specifically, we study robust $(2;1)$ black-box VRF-combiners. Such robust $(2;1)$ black-box VRF-combiners are conceptually the simplest robust black-box VRF-combiner. Still, the existence of a robust $(2;1)$ black-box VRF-combiner immediately implies the existence of a robust $(n;1)$ black-box VRF combiner. On the other hand, an impossibility result implies that there exists no robust $(n;\lfloor n/2 \rfloor)$ black-box VRF-combiner. Consequently, settling the $(2;1)$ case in either direction extends to a seemingly stronger result.

In this section, we prove the impossibility of a robust $(2;1)$ black-box VRF-combiner. At a high level, the proof exploits the fact that the outputs of the candidate schemes must be used to produce a pseudorandom output, yet they cannot influence the output distribution "too much". Otherwise, if a candidate has "too much" influence on the image, it could produce two distinct images such that the verifier cannot detect which image was produced by the honest execution of the evaluation algorithm.

In the proof we make the simplifying assumption that the secret key space of all candidates is given by $\mathcal{SK} = \{0,1\}^{s(\lambda)}$ for a function $s \in \mathsf{poly}(\lambda)$. We state the formal statement in the following theorem.

Theorem 3.5 (Robust (2;1) Black-Box VRF Combiner) *There exists no robust $(2;1)$ black-box combiner (Definition 2.12) for the VRF primitive (Definition 2.9).*

Intuition. Rather than proving the impossibility for robust $(2;1)$ black-box combiners directly, we prove the impossibility for robust $(2;1)$ black-box DAG combiners. The impossibility of a robust $(2;1)$ black-box VRF-combiner then follows directly from the equivalence established in Theorem 3.2.

Now suppose, for a contradiction, that there exists a robust $(2;1)$ black-box DAG VRF-combiner Π_{VRF}. Let $\mathcal{O}_1, \mathcal{O}_2$ be candidate schemes, at least one of which implements a secure VRF. By Definition 2.12, $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ must implement a secure VRF. In particular, it must be perfectly correct and uniquely provable. We derive a contradiction by instantiating $\mathcal{O}_1, \mathcal{O}_2$ with specific candidate schemes at least one of which implements a secure VRF, and then constructing an adversary $\mathcal{A}^{\mathsf{PSPACE},\mathcal{O}_1,\mathcal{O}_2}$ that breaks the pseudorandomness of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ with non-negligible advantage.

The adversary works by iteratively narrowing down the set of candidate secret keys in the pseudorandomness experiment, starting with the secret key space $\mathcal{SK}$, using its PSPACE-complete oracle PSPACE. The filtering works as follows: $\mathcal{A}$ samples a polynomial number of evaluation points $x_1, \ldots, x_{s(\lambda)+\lambda}$ u.a.r. from $\{0,1\}^\lambda$, then uses the evaluation oracle $\mathcal{O}_{\text{eval}}$ in the pseudorandomness security experiment to evaluate these points un-

der the secret key sk^* in the security game to obtain image-proof pairs $(y_1, \pi_1), \ldots, (y_{s(\lambda)+\lambda}, \pi_{s(\lambda)+\lambda})$. Using public key pk^* of the security game, the adversary computes the trace of the combiner's verification algorithm on the tuples $(\mathsf{pk}^*, x_1, y_1, \pi_1), \ldots, (\mathsf{pk}^*, x_{s(\lambda)+\lambda}, y_{s(\lambda)+\lambda}, \pi_{s(\lambda)+\lambda})$ to obtain $\tau_1, \ldots, \tau_{s(\lambda)+\lambda}$. The key ingredient is a ppt. Turing machine $\mathsf{EvalSim}_{\mathsf{VRF}}$ that takes as input a tuple $(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i)$ and returns either $\top$ or $\bot$, satisfying the following three properties:

- $\mathsf{EvalSim}_{\mathsf{VRF}}$ makes no oracle queries to the candidates $\mathcal{O}_1, \mathcal{O}_2$
- If $\mathsf{sk} = \mathsf{sk}^*$ then $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i) = \top$
- If $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}, x_i) \neq y_i$ then $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i) = \bot$

Before elaborating on the construction of $\mathsf{EvalSim}_{\mathsf{VRF}}$, we first explain how the adversary can use this ppt. Turing machine to break the pseudorandomness of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$. Recall that we require the security properties of the combiner to hold relative to a PSPACE-complete oracle precisely because we want the combiner to derive its security from the candidate oracles rather than implementing it itself.

$\mathsf{EvalSim}_{\mathsf{VRF}}$ makes no oracle queries, yet still only returns $\top$ only if a secret key candidate $\mathsf{sk} \in \mathcal{SK}$ *agrees* with the game's secret key sk^* in the sense that on x_i, both evaluate to the same image. Using its PSPACE-complete oracle PSPACE, the adversary can enumerate the entire secret key space $\mathcal{SK}$ and discard any secret key candidate for which there exists a tuple $(x_i, y_i, \pi_i, \tau_i)$ such that $\mathsf{EvalSim}_{\mathsf{VRF}}$ returns $\bot$. (Strictly speaking, the PSPACE oracle decides membership queries for PSPACE languages, we defer these technicalities to the formal proof.) Note that sk^* is never eliminated, since $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}^*, x_i, y_i, \pi_i, \tau_i) = \top$ always holds and all surviving candidates evaluate to the same image as sk^* on $x_1, \ldots, x_{s(\lambda)+\lambda}$. After filtering with every tuple $(x_i, y_i, \pi_i, \tau_i)$, two cases can arise:

If for sufficiently many tuples $(x_i, y_i, \pi_i, \tau_i)$ the number of eliminated candidate secret keys is sufficiently large, then after $s(\lambda) + \lambda$ iterations of filtering, we end up with a polynomially sized set $\mathcal{SK}_{cand}$ containing sk^*. The adversary can then sample a secret key $\mathsf{sk}_{cand} \leftarrow_{\$} \mathcal{SK}_{cand}$ uniformly at random. If $\mathsf{sk}_{cand} = \mathsf{sk}^*$, the adversary wins the pseudorandomness security experiment. This happens with non-negligible probability.

On the other hand, if only for a small number of tuples $(x_i, y_i, \pi_i, \tau_i)$ set of candidate keys decreases significantly in size, then for many of the remaining candidates, the probability of evaluating a uniformly random evaluation point to the same image as sk^* has to be non-negligible. In this scenario, the adversary proceeds analogously: It samples a uniformly random secret key candidate and evaluates the challenge input x^* to compare it with the challenge image y^*. Since sk_{cand} and sk^* evaluate x^* to the same image y^* with

non-negligible probability, the adversary wins the game with non-negligible advantage.

We now describe the construction of the $\mathsf{EvalSim}_{\mathsf{VRF}}$ algorithm. The algorithm simulates the evaluation of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ by iterating over the the vertices in $V(\mathcal{G}_{\mathsf{Eval}})$ of the DAG corresponding to $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}$ in a fixed but arbitrary topological ordering. At every vertex v it simulates the associated Turing machine f_{Eval}^{v} on the corresponding inputs to obtain $\mathbf{q}_v$. Oracle queries to $\mathcal{O}_1$ are answered trivially with a fixed response, this is the case as we will later initialize this oracle with a trivial, uniquely provable and perfectly correct oracle that returns fixed values for key generation and evaluation and verifies exactly the fixed image. For oracle queries to $\mathcal{O}_2$ we distinguish *exposed* queries, which can be read off the verification trace τ_i, and *obscure* queries, which are not contained in the verification trace. The simulator answers the exposed queries with the response contained in τ_i. Thus the only queries that the simulator cannot perfectly answer are the obscure queries to $\mathcal{O}_2$.

For every obscure query to $\mathcal{O}_2$ the adversary branches its execution for every well-formed responses of the query (e.g. all possible values in $\mathcal{Y} \times \mathcal{P}$ if the query is to $\mathcal{O}_2.\mathsf{Eval}_{\mathsf{VRF}}$) and continues the execution for every branch. Finally if for all branches the evaluation of the combiner is y_i, then $\mathsf{EvalSim}_{\mathsf{VRF}}$ returns $\top$, otherwise it returns $\bot$. We illustrate the algorithm $\mathsf{EvalSim}_{\mathsf{VRF}}$ with pseudocode in Figure 3.2.

It is easy to see that the first and third requirements for the $\mathsf{EvalSim}_{\mathsf{VRF}}$ algorithm are met. By construction $\mathsf{EvalSim}_{\mathsf{VRF}}$ makes no oracle queries to candidates. Assuming that $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}, x_i) \neq y_i$ then, since we answer all queries to $\mathcal{O}_1$ and the exposed queries to $\mathcal{O}_2$ truthfully, there must exist a well-formed assignment of oracle responses to obscure queries to candidate $\mathcal{O}_2$ such that the execution of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}, x_i)$ is simulated perfectly and therefore the output is not y_i, thus we have $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i) = \bot$.

For the second property, we argue that for the security experiment's secret key sk^* any assignment of responses to obscure queries to candidate $\mathcal{O}_2$ does not change the image produced by the combiner simulation. Suppose for a contradiction that there exists two different assignments of oracle responses that result in two different images $\tilde{y} \neq \hat{y}$. We derive a contradiction in three steps. First, we may assume without loss of generality that in the evaluation of the robust black-box DAG combiner's evaluation algorithm every specific oracle is queried on a specific input at most once (Claim 3.6). Secondly, we construct two oracles $\widehat{\mathcal{O}}_2, \widetilde{\mathcal{O}}_2$, possibly not uniquely provable or perfectly correct, which are the oracle $\mathcal{O}_2$ remapped on the obscure queries to the two assignments that produce different images $\hat{y} \neq \tilde{y}$ respectively. Third, we turn $\mathcal{O}_1$ into a full VRF by remapping all but a small subset of evaluation points to agree with the secure VRF $\mathcal{O}_2$ (Lemma 3.7). Since $\widehat{\mathcal{O}}_2, \widetilde{\mathcal{O}}_2$ produce

different images but only differ on obscure queries not contained in the verification trace they are indistinguishable for the combiner's verification algorithm. This forces the combiner to violate either unique provability or perfect correctness. We show the description of the full adversary $\mathcal{A}$ in Figure 3.3.

Formal Treatment. Let Π_{VRF} be an arbitrary robust $(2;1)$ black-box DAG combiner for the VRF primitive. Let $\mathcal{O}_{vrf}$ be a VF oracle implementing a secure VRF and let $\mathcal{O}_{uppc}$ be an oracle implementing a trivial, uniquely provable and perfectly correct VF such that the key generation algorithm always returns $(\mathsf{sk}_{def}, \mathsf{pk}_{def})$, the evaluation algorithm returns (y_{def}, π_{def}) for every input and the verification algorithm accepts a tuple (pk, x, y, π) if and only if $y = y_{def}$. First, we show two important results that are central to the proof stated in Claim 3.6 and Lemma 3.7.

We claim that we can assume without loss of generality, that the robust black-box DAG combiner for the VRF primitive, during the execution of its evaluation algorithm on arbitrary inputs, queries every oracle algorithm on any given input at most once.

Claim 3.6 *Let Π_{VRF} be a robust $(n;k)$ black-box DAG VRF-combiner (Definition 3.1). There exists a robust $(n;k)$ black-box DAG VRF-combiner $\widetilde{\Pi}_{\mathsf{VRF}}$ such that in the execution of the evaluation algorithm, any oracle algorithm is queried on any given input at most once.*

Proof (Claim 3.6) Let Π_{VRF} be a robust $(n;k)$ black-box DAG VRF-combiner. We obtain $\widetilde{\Pi}_{\mathsf{VRF}}$ by modifying Π_{VRF} in the following way: At every vertex $v \in V(\mathcal{G}_{\mathrm{Eval}})$ the Turing machine $\tilde{f}^v_{\mathrm{Eval}}$ maintains a table T_v of mappings from previous oracle queries $\mathbf{q}_w$ for $w \in \mathcal{N}^-_{\mathcal{G}_{\mathrm{Eval}}}(v)$ to oracle responses $\mathbf{x}_w$ of the same oracle algorithm $\ell_{eval}(v)$. $\widetilde{f}^v_{eval}$ then simulates the Turing machine f^v_{Eval} until it returns the query $\mathbf{q}_v$. Then $\widetilde{f}^v_{\mathrm{Eval}}$ checks whether $\mathbf{q}_w$ is contained in the table T_v. If that is not the case we issue the query $\mathbf{q}_w$ to $\mathcal{O}_{\ell_{\mathrm{Eval}}}(\mathbf{q}_w)$, otherwise we issue a fresh query $\widetilde{\mathbf{q}}_w$ not in the table. If the oracle is queried on $\widetilde{\mathbf{q}}_w$ subsequent Turing machines f^u_{Eval} with $(v,u) \in E(\mathcal{G}_{\mathrm{Eval}})$, will then simply find the first query to the oracle $\ell_{\mathrm{Eval}}(v)$ with the same input if they are also incident to u. To ensure this, we will again simply draw an edge from the source to every other vertex, from every vertex (d,i,j) to any vertex (d',i',j') if $d < d'$ and from every vertex to the sink.

To show that $\widetilde{\Pi}_{\mathsf{VRF}}$ is indeed a valid robust $(n;k)$ black-box DAG VRF-combiner it is sufficient to show that for any tuple of n candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$ of which at least k implement a secure VRF the algorithms implemented by $\Pi^{\mathcal{O}_1,\ldots,\mathcal{O}_n}_{\mathsf{VRF}}$ and $\widetilde{\Pi}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}_{\mathsf{VRF}}$ have equivalent output distributions. Since we do not modify the algorithms $\mathsf{Gen}_{\mathsf{VRF}}$ and $\mathsf{Vfy}_{\mathsf{VRF}}$, we only need to show that equivalence holds for $\mathsf{Eval}_{\mathsf{VRF}}$.

Assume towards a contradiction that this is not the case, i.e. that there exists an input $(\mathsf{sk}, x) \in \mathcal{SK} \times \{0,1\}^\lambda$ such that $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) = y_1$ and $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) = y_2$ with $y_1 \neq y_2$. Since the Turing machines around the oracle calls are simulated the same way in both, this either means that f_{eval}^t is probabilistic or that an oracle query is answered differently and thus not deterministic. In both cases, both y_1 and y_2 could also have been produced by $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$. By perfect correctness both y_1, y_2 must be verifiable but by unique provability only at most one of y_1, y_2 can be verifiable. A contradiction. We conclude that therefore the algorithm $\mathsf{Eval}_{\mathsf{VRF}}$ of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ and $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ are functionally equivalent.

It remains to show that $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ runs in polynomial time. Observe, that the number of vertices in $\mathcal{G}_{\mathrm{Eval}}$ remains unchanged from $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$. Therefore, $|V(G_{\mathrm{Eval}})| \in \mathsf{poly}(n, \lambda)$. At each vertex v, the Turing machine $\widetilde{f}_{\mathrm{Eval}}^v$ performs the same computation as f_{Eval}^v plus a table lookup in T_v. Since the size of the table is bounded from above by the number of vertices $|V(G_{\mathrm{Eval}})|$ the lookup is efficiently computable. Hence, the overall running time of $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$ is bounded by a polynomial in n and λ. □

From now on we will assume that for every input to the combiner's evaluation algorithm, an oracle is queried on any given input at most once. That is, if we remap the output of an oracle on a single point, then at most one oracle query differs in the execution of the evaluation algorithm for every input.

Next, let $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ and $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ be sets with size bounded by a polynomial, i.e. $|\widetilde{\mathcal{SK}}|, |\widetilde{\mathcal{PK}}| \in \mathsf{poly}(\lambda)$. We can construct an oracle $\widetilde{\mathcal{O}}_{uppc}$ from $\mathcal{O}_{vrf}$ and $\mathcal{O}_{uppc}$ such that $\widetilde{\mathcal{O}}_{uppc}$ implements a secure VRF and $\widetilde{\mathcal{O}}_{uppc}$ agrees with $\mathcal{O}_{uppc}$ on the outputs of the evaluation algorithm on all points $\widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}$ and on the verification algorithm on all points $\widetilde{\mathsf{pk}} \in \widetilde{\mathcal{PK}}$. Formally stated in the following lemma.

Lemma 3.7 *Let $\mathcal{O}_{vrf}$ be a VRF oracle (Definition 2.9) and $\mathcal{O}_{uppc}$ a VF oracle that is perfectly correct (Definition 2.4) and uniquely provable (Definition 2.5). Let $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ and $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ be polynomially bounded sets, i.e. $|\widetilde{\mathcal{SK}}|, |\widetilde{\mathcal{PK}}| \in \mathsf{poly}(\lambda)$. There exists an oracle $\widetilde{\mathcal{O}}_{uppc}$ satisfying the following properties.*

1. *The evaluation algorithm of $\widetilde{\mathcal{O}}_{uppc}$ agrees with the evaluation algorithm of $\mathcal{O}_{uppc}$ on all secret keys $\widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}$, i.e. we have that*

$$\forall \widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}, \widetilde{\mathcal{O}}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, \cdot) = \mathcal{O}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, \cdot)$$

2. *The verification algorithm of $\widetilde{\mathcal{O}}_{uppc}$ agrees with the verification algorithm of $\mathcal{O}_{uppc}$ on all public keys $\widetilde{\mathsf{pk}} \in \widetilde{\mathcal{PK}}$, i.e. we have that*

$$\forall \widetilde{\mathsf{pk}} \in \widetilde{\mathcal{PK}}, \widetilde{\mathcal{O}}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, \cdot, \cdot, \cdot) = \mathcal{O}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, \cdot, \cdot, \cdot)$$

3. *The oracle $\widetilde{\mathcal{O}}_{uppc}$ implements a secure VRF (Definition 2.9).*

A naive idea would be to replace the mapping of $\mathcal{O}_{vrf}.\mathsf{Eval}_{\mathsf{VRF}}$ on all inputs (sk, x) with $\mathsf{sk} \in \widetilde{\mathcal{SK}}$ and similarly modify the mapping of $\mathcal{O}_{vrf}.\mathsf{Vfy}_{\mathsf{VRF}}$ on all inputs (pk, x, y, π) with $\mathsf{pk} \in \widetilde{\mathcal{PK}}$. However, the resulting oracle doesn't necessarily need to be uniquely provable or perfectly correct anymore. For instance, imagine the scenario where a secret key $\widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}$ is remapped but the corresponding public key pk does not itself belong to $\widetilde{\mathcal{PK}}$. In that case the verification algorithm on pk is left unchanged while the evaluation algorithm on $\widetilde{\mathsf{sk}}$ is modified. This can result in correctly computed image-proof pairs to be rejected by the verification algorithm, violating perfect correctness.

We begin by introducing two relations $\sim_{\mathcal{SK}} \subseteq \mathcal{SK}^2$ and $\sim_{\mathcal{PK}} \subseteq \mathcal{PK}^2$. Intuitively, these relations capture whether two secret or public keys have a counterpart which could have been sampled with both keys. Formally, we define the relations $\sim_{\mathcal{SK}}, \sim_{\mathcal{PK}}$ as follows:

$$\mathsf{sk}_1 \sim_{\mathcal{SK}} \mathsf{sk}_2 \iff \exists \mathsf{pk} \in \mathcal{PK}, \Pr\left[\begin{matrix}(\mathsf{sk}_1, \mathsf{pk}) \leftarrow_{\$} \mathcal{O}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda) \wedge \\ (\mathsf{sk}_2, \mathsf{pk}) \leftarrow_{\$} \mathcal{O}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)\end{matrix}\right] > 0 \tag{3.5}$$

$$\mathsf{pk}_1 \sim_{\mathcal{PK}} \mathsf{pk}_2 \iff \exists \mathsf{sk} \in \mathcal{SK}, \Pr\left[\begin{matrix}(\mathsf{sk}, \mathsf{pk}_1) \leftarrow_{\$} \mathcal{O}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda) \wedge \\ (\mathsf{sk}, \mathsf{pk}_2) \leftarrow_{\$} \mathcal{O}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)\end{matrix}\right] > 0 \tag{3.6}$$

We now consider the reflexive transitive closure of those relations $\sim^*_{\mathcal{SK}}, \sim^*_{\mathcal{PK}}$, formally defined by $\sim^* = \bigcup_{k \in \mathbb{N}} \sim^k$. Since $\sim_{\mathcal{SK}}$ and $\sim_{\mathcal{PK}}$ are symmetric by definition, their reflexive transitive closures are reflexive, symmetric and transitive and hence equivalence relations. Given an element sk, we denote its equivalence class under $\sim^*_{\mathcal{SK}}$ with $[\mathsf{sk}]_{\sim^*_{\mathcal{SK}}} = \{\mathsf{sk}' \in \mathcal{SK} : \mathsf{sk}' \sim^*_{\mathcal{SK}} \mathsf{sk}\}$, analogously for an element pk and the relation $\sim^*_{\mathcal{PK}}$ we define $[\mathsf{pk}]_{\sim^*_{\mathcal{PK}}} = \{\mathsf{pk}' \in \mathcal{PK} : \mathsf{pk}' \sim^*_{\mathcal{PK}} \mathsf{pk}\}$.

Additionally, for secret and public keys we define the function $R_{\mathcal{SK}}$ and $R_{\mathcal{PK}}$ respectively that map a key to all counterparts which are sampled with that key with non-negligible probability.

$$R_{\mathcal{PK}}(\mathsf{pk}) = \left\{\mathsf{sk} \in \mathcal{SK} : \Pr\left[(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)\right] > 0\right\} \tag{3.7}$$

$$R_{\mathcal{SK}}(\mathsf{sk}) = \left\{\mathsf{pk} \in \mathcal{PK} : \Pr\left[(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)\right] > 0\right\} \tag{3.8}$$

Next, we define the notion of the closure of a set of secret or public keys. Given a subset of secret keys $\widetilde{\mathcal{SK}}$ or public keys $\widetilde{\mathcal{PK}}$, we define its closure $\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})$ and $\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})$ which are just the union over the equivalence classes

of all keys in the subsets. We formally define the functions as follows:

$$\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}) = \bigcup_{\widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}} [\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}} \tag{3.9}$$

$$\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}}) = \bigcup_{\widetilde{\mathsf{pk}} \in \widetilde{\mathcal{PK}}} [\widetilde{\mathsf{pk}}]_{\sim^*_{\mathcal{PK}}} \tag{3.10}$$

Observe, that the secret key closure $\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})$ for a subset $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ is the set of secret keys that need to be remapped if we want to remap all of the evaluations on the secret keys $\widetilde{\mathcal{SK}}$ to preserve unique provability and correctness. The same holds for the public key closure. Now we define the closure for subsets $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ and $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ at the same time. They correspond to the minimal set of secret or public keys that need to be remapped to preserved unique provability and perfect correctness of $\mathcal{O}_{uppc}$ if we want to remap all keys in $\widetilde{\mathcal{SK}}$ and $\widetilde{\mathcal{PK}}$.

$$\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) = \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}) \cup \bigcup_{\widetilde{\mathsf{pk}} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} R_{\mathcal{PK}}(\widetilde{\mathsf{pk}}) \tag{3.11}$$

$$\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) = \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}}) \cup \bigcup_{\widetilde{\mathsf{sk}} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})} R_{\mathcal{SK}}(\widetilde{\mathsf{sk}}) \tag{3.12}$$

Finally, we construct an oracle $\widetilde{\mathcal{O}}_{uppc}$ that interpolates between $\mathcal{O}_{uppc}$ and $\mathcal{O}_{vrf}$. The key generation algorithm is defined to be the function for key generation of $\mathcal{O}_{vrf}$, the evaluation and the verification algorithm of $\widetilde{\mathcal{O}}_{uppc}$ are defined as follows: They evaluate to the output of $\mathcal{O}_{uppc}$ on all keys in $\widetilde{\mathcal{SK}}$ and $\widetilde{\mathcal{PK}}$ and they evaluate to the output of $\mathcal{O}_{vrf}$ otherwise.

$$\widetilde{\mathcal{O}}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) = \begin{cases} \mathcal{O}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) & \text{if } \mathsf{sk} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) \\ \mathcal{O}_{vrf}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) & \text{otherwise} \end{cases} \tag{3.13}$$

$$\widetilde{\mathcal{O}}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi) = \begin{cases} \mathcal{O}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi) & \text{if } \mathsf{pk} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) \\ \mathcal{O}_{vrf}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi) & \text{otherwise} \end{cases} \tag{3.14}$$

The properties 1. and 2. of Lemma 3.7 hold by definition of $\widetilde{\mathcal{O}}_{uppc}$ since $\widetilde{\mathcal{SK}} \subseteq \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}) \subseteq \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$ and $\widetilde{\mathcal{PK}} \subseteq \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}}) \subseteq \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$ which hold by reflexivity of $\sim^*_{\mathcal{SK}}$ and $\sim^*_{\mathcal{PK}}$ respectively. We show that 3. holds by showing that $\widetilde{\mathcal{O}}_{uppc}$ is uniquely provable, perfectly correct and pseudorandom.

Claim 3.8 *Let $\mathcal{O}_{uppc}$ be a VF oracle that is uniquely provable (Definition 2.5) and perfectly correct (Definition 2.4) and let $\mathcal{O}_{vrf}$ be a VF oracle that implements a secure VRF (Definition 2.9). The oracle $\widetilde{\mathcal{O}}_{uppc}$ constructed from $\mathcal{O}_{up}$ and $\mathcal{O}_{vrf}$ for*

given subsets $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ *and* $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ *as in* (3.13) *and* (3.14) *is perfectly correct (Definition 2.4) and uniquely provable (Definition 2.5).*

Before we prove Claim 3.8 we claim that if a secret key sk can be sampled together with a public key pk, for any subset of secret keys $\widetilde{\mathcal{SK}}$ and any subset of public keys $\widetilde{\mathcal{PK}}$, sk is in the closure of $\widetilde{\mathcal{SK}}$ if and only if pk is in the closure of $\widetilde{\mathcal{PK}}$. Since the claim is intuitive and the proof is long we defer the proof to the appendix in Chapter 4.

Claim 3.9 *Let* $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \widetilde{\mathcal{O}}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$ *and* $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ *and* $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ *be arbitrary subsets. Then the following holds:*

$$\mathsf{sk} \in cl_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) \iff \mathsf{pk} \in cl_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$$

Proof (Claim 3.8) Let $\lambda \in \mathbb{N}$ be arbitrary but fixed.

Correctness: Let $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \widetilde{\mathcal{O}}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$, let $x \in \{0,1\}^\lambda$ be arbitrary and let $(y, \pi) \leftarrow \widetilde{\mathcal{O}}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$. If $\mathsf{sk} \in \widetilde{\mathcal{SK}}$ then by Claim 3.9 $\mathsf{pk} \in \widetilde{\mathcal{PK}}$ and correctness holds by the correctness of $\mathcal{O}_{uppc}$. Otherwise, if $\mathsf{sk} \notin \widetilde{\mathcal{SK}}$ then by Claim 3.9 $\mathsf{pk} \notin \widetilde{\mathcal{PK}}$ and correctness holds by the correctness of $\mathcal{O}_{vrf}$.

Unique Provability: Assume towards a contradiction that there exist two image-proof pairs (y_1, π_1) and (y_2, π_2) with distinct images $y_1 \neq y_2$ such that $\widetilde{\mathcal{O}}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y_1, \pi_1) = 1$ and $\widetilde{\mathcal{O}}_{uppc}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y_2, \pi_2) = 1$. If $\mathsf{pk} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$ then verification is defined by $\mathcal{O}_{uppc}$ and we arrive at a contradiction with the unique provability of $\mathcal{O}_{uppc}$. If $\mathsf{pk} \notin \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$ then verification is defined by $\mathcal{O}_{vrf}$ and therefore we arrive at a contradiction with the assumed unique provability of $\mathcal{O}_{vrf}$. In either case we reach a contradiction, hence there cannot exist such image-proof pairs. □

Claim 3.10 *Let* $\mathcal{O}_{uppc}$ *be a VF oracle that is uniquely provable (Definition 2.5) and perfectly correct (Definition 2.4) and let* $\mathcal{O}_{vrf}$ *be a VF oracle that implements a secure VRF (Definition 2.9) that is* $(\mathsf{t}^{prf}_{vrf}, \varepsilon^{prf}_{vrf})$*-pseudorandom (Definition 2.6). The oracle* $\widetilde{\mathcal{O}}_{uppc}$ *constructed from* $\mathcal{O}_{up}$ *and* $\mathcal{O}_{vrf}$ *for given subsets* $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ *and* $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ *as in* (3.13) *and* (3.14) *is* $(\mathsf{t}^{prf}_{uppc}, \varepsilon^{prf}_{uppc})$*-pseudorandom (Definition 2.6). Where*

$$\varepsilon^{prf}_{uppc} \leq \varepsilon^{prf}_{vrf} + \mu \quad \textit{and} \quad \mathsf{t}^{prf}_{vrf} = \mathsf{t}^{prf}_{uppc}$$

for $\mu \in \mathsf{negl}(\lambda)$*. Therefore, if* $\mathcal{O}_{vrf}$ *is pseudorandom, then* $\widetilde{\mathcal{O}}_{uppc}$ *is pseudorandom.*

We prove the claim with a hybrid argument. Intuitively the proof works by bounding the probability that a bad event $\mathcal{E}$: that the secret key sk^* sampled in the pseudorandomness security experiment is in the set $\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$. If $\mathcal{E}$ doesn't occur, then the security game is identical to the security game against $\mathcal{O}_{vrf}$ which no adversary running in time t^{prf}_{vrf} wins with advantage greater

than ε^{prf}_{vrf}. Using the difference lemma we can then bound the advantage of any adversary playing the pseudorandomness game against $\widetilde{\mathcal{O}}_{uppc}$ from above by an expression in ε^{prf}_{vrf}.

We define games H_1, H_2, H_3 as follows.

- H_1 is the simple pseudorandomness experiment for the hidden bit $b = 0$ against the oracle $\mathcal{O}_{vrf}$, i.e. the adversary is playing against the prf security game where he is given a challenge image y^* computed from the challenge pre-image x^* using $\mathcal{O}_{vrf}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*, x^*)$ where sk^* is the hidden secret key sampled by the challenger.
- H_2 is the simple pseudorandomness experiment for the hidden bit $b = 0$ against the oracle $\widetilde{\mathcal{O}}_{uppc}$, i.e. the adversary is playing the prf security game where he is given a challenge image y^* computed from the challenge pre-image x^* using $\widetilde{\mathcal{O}}_{uppc}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*, x^*)$ where sk^* is the hidden secret key sampled by the challenger.
- H_3 is the simple pseudorandomness experiment for the hidden bit $b = 1$ against the oracle $\mathcal{O}_{vrf}$, i.e. the adversary is playing the prf security game where he is given a challenge image y^* which the challenger sampled u.a.r. from the image codomain $\mathcal{Y}$.

We write $H_i^{\lambda,\mathcal{A}} = 1$ to denote the event that adversary $\mathcal{A}$ guesses bit $b' = 1$ in the game H_i for all $i \in [3]$. We can now bound the advantage of any adversary $\mathcal{A}$ against $\widetilde{\mathcal{O}}_{uppc}$ in the pseudorandomness security experiment as follows. We make the following three claims whose proof we defer to the appendix:

Claim 3.11 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \right] - \frac{1}{2} \right| = \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right|$$

Claim 3.12 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$\left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_1^{\lambda,\mathcal{A}} = 1 \right] \right| \leq \varepsilon^{prf}_{vrf}$$

Claim 3.13 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$\left| \Pr\left[H_1^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right| \leq \mu(\lambda)$$

Where $\mu \in \mathsf{negl}(\lambda)$.

Proof (Claim 3.10) Using Definition 2.6 for (3.15), Claim 3.11 for (3.16), Claim 3.12 for (3.19) and Claim 3.13 for (3.20) we obtain the desired bound. (3.17) is the result of subtracting and adding $\Pr[H_1^{\lambda,\mathcal{A}} = 1]$ and then applying the triangle inequality.

$$\begin{aligned}
\mathbf{Adv}^{\mathrm{prf}}_{\tilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) &= 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\tilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \right] - \frac{1}{2} \right| && (3.15)\\
&= \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right| && (3.16)\\
&\leq \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_1^{\lambda,\mathcal{A}} = 1 \right] \right| && (3.17)\\
&\quad + \left| \Pr\left[H_1^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right| && (3.18)\\
&\leq \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_1^{\lambda,\mathcal{A}} = 1 \right] \right| + \varepsilon^{prf}_{vrf} && (3.19)\\
&\leq \mu + \varepsilon^{prf}_{vrf} && (3.20)
\end{aligned}$$

□

Proof (Lemma 3.7) The claim follows from Claim 3.8 and Claim 3.10. □

Equipped with Lemma 3.7, we now consider the combiner Π_{VRF} initialized with the oracles $\mathcal{O}_{vrf}$ and $\mathcal{O}_{up}$ as introduced above. We compute:

$$\begin{aligned}
x &\leftarrow_{\$} \{0,1\}^{\lambda}\\
(\mathsf{sk}, \mathsf{pk}) &\leftarrow_{\$} \Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}.\mathsf{Gen}_{\mathsf{VRF}}(1^{\lambda})\\
(y, \pi) &\leftarrow \Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf}\mathcal{O}_{up}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)\\
\tau &\leftarrow \mathsf{trace}\Big(\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)\Big)
\end{aligned}$$

We classify the vertices of $V(\mathcal{G}_{\mathrm{Eval}})$ as follows. A vertex $v \in V(G_{\mathrm{Eval}})$ is called *exposed* if the oracle query issued at v during the evaluation of the input (sk, x) also appears in the verification trace τ. Otherwise, we call v *obscure*. Observe, that exposed queries and queries to $\mathcal{O}_{uppc}$ can be answered without calling any oracle by looking at the verification trace or taking the default values $\mathsf{sk}_{def}, \mathsf{pk}_{def}, y_{def}, \pi_{def}, b_{def}$ respectively (where b_{def} depends on the input y to the verification). The only queries whose responses are not directly available without any oracle queries are the obscure queries to $\mathcal{O}_{vrf}$. We claim that these responses can be substituted with arbitrary, but well-formed, oracle responses without affecting the combiners output (sk, x).

Claim 3.14 *Let $\mathcal{O}_{vrf}$ and $\mathcal{O}_{uppc}$ be defined as above, let $\mathsf{sk}, \mathsf{pk}, x, y, \pi, \tau$ be defined as above and let $V_{obs} \subseteq V(\mathcal{G}_{Eval})$ be the set of obscure vertices to $\mathcal{O}_{vrf}$. Then the oracles responses at vertices in V_{obs} are irrelevant in the sense that for any assignment of well-formed responses to the queries at vertices in V_{obs} the algorithm $\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}.\mathsf{Eval}_{\mathsf{VRF}}$ produces the same image y.*

Proof (Claim 3.14) Assume this were not the case, let $\widehat{\mathcal{O}}_{vrf}$ and $\widetilde{\mathcal{O}}_{vrf}$ be the oracles obtained from $\mathcal{O}_{vrf}$ by modifying the mapping on the obscure vertices V_{obs} to the two assignments to obtain two different images $y_0 \neq y_1$ and let $\widetilde{\mathcal{O}}_{uppc}$ be the VRF oracle obtained from $\mathcal{O}_{uppc}$ by applying the procedure detailed in Lemma 3.7. Since $\widetilde{\mathcal{O}}_{uppc}$ is a secure VRF, the combiner initialized with the oracles $\widehat{\mathcal{O}}_{vrf}, \widetilde{\mathcal{O}}_{uppc}$ and the combiner initialized with the oracle $\widetilde{\mathcal{O}}_{vrf}, \widetilde{\mathcal{O}}_{uppc}$ must implement secure VRFs and in particular also uniquely provable and perfectly correct VFs. Further note that by Claim 3.6 we note that there can be no conflict of having to assign two different outputs to the query made to the same oracle on the same input at a different position since every oracle algorithm is queried at any input at most once. Hence:

$$\Pi_{\mathsf{VRF}}^{\widetilde{\mathcal{O}}_{vrf},\widetilde{\mathcal{O}}_{uppc}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, \tilde{y}, \tilde{\pi}) = 1 \tag{3.21}$$

$$\Pi_{\mathsf{VRF}}^{\widehat{\mathcal{O}}_{vrf},\widetilde{\mathcal{O}}_{uppc}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, \tilde{y}, \tilde{\pi}) = 0 \tag{3.22}$$

Since the oracle $\widehat{\mathcal{O}}_{vrf}$ and $\widetilde{\mathcal{O}}_{vrf}$ only differ on obscure queries we have that the verification algorithm on $(\mathsf{pk}, x, \tilde{y}, \tilde{\pi})$ must evaluate the same and to the same result. This is a contradiction with (3.21) and (3.22). □

We define the evaluation simulation $\mathsf{EvalSim}_{\mathsf{VRF}}$ that simulates the evaluation without invoking any oracle queries. The simulation will simply traverse the evaluation algorithm of the DAG combiner. For any oracle query it checks the corresponding transcript τ_i. If the query is either to oracle $\mathcal{O}_1$ we can honestly answer it with $\mathbf{x}_v^{def}$ or in the transcript, then we can answer it honestly as well. If the query is *obscure* then we branch once for every possible assignment of the oracle response which fits the given format, e.g. for a query to eval we branch for every answer $(y, \pi) \in \mathcal{Y} \times \mathcal{P}$. The pseudocode for $\mathsf{EvalSim}_{\mathsf{VRF}}$ is shown in Figure 3.2. In the following we show that this algorithm satisfies a few properties. We want to note that the evaluation simulation algorithm is clearly inefficient since it enumerates possibly many well-formed oracle queries, since the adversary will use it only in combination with the PSPACE-complete oracle in the end that is not a problem.

Algorithm $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau)$

$v_1, \ldots, v_m \leftarrow$ topological ordering of $V(\mathcal{G}_{eval}) \setminus \{s_{eval}, t_{eval}\}$
$c_{s_{eval}} \leftarrow 1$ // Number of states or branches
for $i = 1$ to m **do**
  // Reconstruct all branches:
  $\Sigma \leftarrow \varnothing$
  for $w \in \mathcal{N}^-_{\mathcal{G}_{eval}}(v_i)$ **do**
    $\Sigma \leftarrow \{\Sigma \cup \{(w, \mathbf{x}_w^{(j)})\} : j \in [c_w]\}$
  // Iterate all branches:
  $j \leftarrow 1$
  for $\sigma \in \Sigma$ **do**
    $\mathbf{q}_{v_i}^{(j)} \leftarrow f_{eval}^{v_i}(\sigma)$
    if $\ell(v_i) = (2, _)$ **then** // Query to oracle $\mathcal{O}_{uppc}$
      $\mathbf{x}_{v_i}^{(j)} \leftarrow \mathbf{x}_{v_i}^{def}$
    else if $\mathbf{q}_{v_i}^{(j)} \in \tau$ **then** // Exposed query to $\mathcal{O}_{vrf}$
      $\mathbf{x}_{v_i}^{(j)} \leftarrow$ response to $\mathbf{q}_{v_i}^{(j)}$ in τ
    else if $\ell(v_i) = (1, _)$ **then** // Obscure query to $\mathcal{O}_{vrf}$
      for $\mathbf{s} \in \mathcal{S}_{v_i}$ **do**
        $\mathbf{x}_{v_i}^{(j)} \leftarrow \mathbf{s}$
        $j \leftarrow j + 1$
    $c_{v_i} \leftarrow j$
// Check if all evaluation branches result in y
  $\Sigma \leftarrow \varnothing$
  for $w \in \mathcal{N}^-_{\mathcal{G}_{eval}}(t_{eval})$ **do**
    $\Sigma \leftarrow \{\Sigma \cup \{(w, \mathbf{x}_w^{(j)})\} : j \in [c_w]\}$
  for $\sigma \in \Sigma$ **do**
    $\mathbf{y}_{t_{eval}} \leftarrow f_{eval}^{t_{eval}}(\sigma)$
    if $\mathbf{y}_{t_{eval}} \neq y$ **then**
      return $\bot$
return $\top$

Figure 3.2: Pseudocode of the algorithm $\mathsf{EvalSim}_{\mathsf{VRF}}$. We use c_v to denote the number of possible branches at vertex v, $\mathbf{x}_v^{def}$ for the default value of the oracle of $\mathcal{O}_{uppc}$ at vertex v (e.g. if $c(v) = (2,2)$ then (y_{def}, π_{def})) and with $\mathcal{S}_v$ the corresponding codomain of the oracle at v (e.g. if $c(v) = (1,2)$ then $\mathcal{Y} \times \mathcal{P}$).

Claim 3.15 *Let* $\mathsf{EvalSim}_{\mathsf{VRF}}$ *be the algorithm as described above and in Figure 3.2 taking as input a tuple* $(\mathsf{sk}, x, y, \pi, \tau)$ *containing a secret key, evaluation point, image, proof and a trace of the combiner's verification algorithm on* (pk, x, y, π) *where* pk *is the true public key. Then* $\mathsf{EvalSim}_{\mathsf{VRF}}$ *satisfies the following properties.*

1. $\mathsf{EvalSim}_{\mathsf{VRF}}$ *makes no oracle queries to the candidates* $\mathcal{O}_{vrf}, \mathcal{O}_{up}$
2. *If* $\mathsf{sk} = \mathsf{sk}^*$ *then* $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau) = \top$
3. *If* $\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}, x) \neq y$ *then* $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau) = \bot$

Informally, this means that the evaluation simulation will return with $\top$ on a non-empty subset of secret keys which agrees with the true secret key sk^* on the evaluation of x. Non-empty because the security experiment's secret key sk^* will always result in $\top$ due to the second property.

Proof (Claim 3.15) The first property holds by construction. The second property follows directly from Claim 3.14. The third property holds because the true execution of the combiner's evaluation algorithm corresponds is always a branch in the execution of $\mathsf{EvalSim}_{\mathsf{VRF}}$, if the output of that branch is not y then by construction the combiner will return $\bot$. □

Using this simulator $\mathsf{EvalSim}_{\mathsf{VRF}}$ we can describe the set of iteratively filtered candidate secret keys as follows. Let $\{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]}$ be a set of $s(\lambda) + \lambda$ tuples containing a uniformly random evaluation point $x_i \in \{0,1\}^\lambda$, the evaluation of x_i resulting in the tuple (y_i, π_i) and the trace of the combiner's verification algorithm on (x_i, y_i, π_i). The set of candidate keys is defined as:

$$\mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i) := \{\mathsf{sk} \in \mathcal{SK} : \mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i)\}$$

We claim that after filtering for $s(\lambda) + \lambda$ rounds, the probability that a remaining candidate key sk_{cand} sampled u.a.r. from the set of remaining candidate keys evaluates a uniformly random evaluation point $x^* \in \{0,1\}^*$ to the same image y^* as the game's secret key sk^* does is non-negligible.

Claim 3.16 *Let* $\mathcal{SK}_{cand}$ *be defined as above and let* $\{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]}$ *then we have for any* sk^* *sampled u.a.r. from the set of filtered candidate keys, i.e.*

$$\mathsf{sk}^* \leftarrow_{\$} \bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i)$$

that the probability of a uniformly random challenge pre-image $x^* \in \{0,1\}^\lambda \setminus \{x_1, \ldots, x_{s(\lambda)+\lambda}\}$ *the black-box combiner instanciated with the oracles* $\mathcal{O}_{vrf}, \mathcal{O}_{uppc}$ *evaluates to the same image* y^* *as for the secret key* sk^* *is non-negligible, i.e.*

$$\Pr\left[\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}^*, x) = \Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}, x)\right] \geq \frac{1}{2} - \frac{1}{2^{\lambda+1}}$$

Proof (Claim 3.16) For a secret key $\mathsf{sk} \in \mathcal{SK}$ we define $\mathcal{X}(\mathsf{sk})$ to be the number of evaluation points $x \in \{0,1\}^\lambda$ for which the evaluation of x under sk agrees with the evaluation of x under sk^*, the game's secret key, i.e.

$$\mathcal{X}(\mathsf{sk}) = \left|\left\{x \in \{0,1\}^\lambda : \Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}, x) = \Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*, x)\right\}\right|$$

We say that a secret key sk is *unfavorable*, if the probability of an evaluation point x sampled u.a.r. from $\{0,1\}^\lambda$ evaluating to a different image under sk then under sk^* is at least $1/2$. Let $\mathcal{SK}_{unfav} \subseteq \mathcal{SK}$ be the set of unfavorable secret keys. Note that the probability of a secret key sk surviving a filtering step is given by the following.

$$\Pr[\mathsf{sk} \in \mathcal{SK}_{cand}(x, y, \pi, \tau)] = \frac{\mathcal{X}(\mathsf{sk})}{2^\lambda}$$

We can bound the expected number of unfavorable secret keys that survive $s(\lambda) + \lambda$ rounds of secret key filtering over the randomness of $x_1, \ldots, x_{s(\lambda)+\lambda}$. We use $\mathbf{1}_{\mathcal{E}}$ to denote the random indicator variable for the event $\mathcal{E}$.

$$\mathbb{E}\left[\left|\mathcal{SK}_{unfav} \cap \bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i)\right|\right] \tag{3.23}$$

$$= \mathbb{E}\left[\sum_{\mathsf{sk}_{unfav}\in\mathcal{SK}_{unfav}} \mathbf{1}_{\mathsf{sk}_{unfav}\in\bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_i,y_i,\pi_i,\tau_i)}\right] \tag{3.24}$$

$$= \sum_{\mathsf{sk}_{unvaf}\in\mathcal{SK}_{unfav}} \Pr\left[\mathsf{sk}_{unfav} \in \bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i)\right] \tag{3.25}$$

$$\leq \sum_{\mathsf{sk}_{unvaf}\in\mathcal{SK}_{unfav}} \prod_{i\in[s(\lambda)+\lambda]} \Pr\left[\mathsf{sk}_{unfav} \in \mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i)\right] \tag{3.26}$$

$$\leq \sum_{\mathsf{sk}_{unvaf}\in\mathcal{SK}_{unfav}} \prod_{i\in[s(\lambda)+\lambda]} \frac{\mathcal{X}(\mathsf{sk}_{unfav})}{2^\lambda} \tag{3.27}$$

$$\leq \sum_{\mathsf{sk}_{unvaf}\in\mathcal{SK}_{unfav}} \prod_{i\in[s(\lambda)+\lambda]} \frac{1}{2} \tag{3.28}$$

$$\leq \frac{1}{2^\lambda} \tag{3.29}$$

Where in (3.25) we use linearity of expectation and the fact that $\mathbf{1}_{\mathcal{E}}$ are Bernoulli random variables and in (3.26) we use that the evaluation points $x_1, \ldots, x_{s(\lambda)+\lambda}$ are sampled i.i.d. from $\{0,1\}^\lambda$. We can now use a simple Markov bound to show that the probability of a unfavorable key surviving the filtering is negligibly small. Note that the expected number of surviving

unfavorable secret keys is by definition non-negative and therefore we have that the probability of all keys filtered is given by the following expression:

$$\Pr\left[\left|\mathcal{SK}_{unfav} \cap \bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{card}(x_i, y_i, \pi_i, \tau_i)\right| \geq 1\right] \leq \frac{1}{2^\lambda}$$

We define the following events:

$$\mathcal{F} := \Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}^*, x^*) = \Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}.\mathsf{Eval}_{\mathsf{VRF}}^{y}(\mathsf{sk}, x^*) \tag{3.30}$$

$$\mathcal{E} := \left|\mathcal{SK}_{unfav} \cap \textstyle\bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{card}(x_i, y_i, \pi_i, \tau_i)\right| \geq 1 \tag{3.31}$$

Note that $\mathcal{F}$ is exactly the event whose probability we want to bound for Claim 3.16 and $\mathcal{E}$ is the event that at least one unfavorable key survives the filtering. Therefore, $\neg\mathcal{E}$ is the event that only favorable keys survive the filtering. To bound $\Pr[\mathcal{F}]$, we condition on the event $\neg\mathcal{E}$ using the fact that a favorable key has probability at least $\frac{1}{2}$ to evaluate to the same image y^* on a uniformly random evaluation point as sk^*.

$$\begin{aligned}
\Pr[\mathcal{F}] &= \Pr[\mathcal{F} \mid \neg\mathcal{E}]\Pr[\neg\mathcal{E}] + \Pr[\mathcal{F} \mid \mathcal{E}]\Pr[\mathcal{E}] && (3.32)\\
&\geq \Pr[\mathcal{F} \mid \neg\mathcal{E}]\Pr[\neg\mathcal{E}] && (3.33)\\
&\geq \tfrac{1}{2}(1 - \tfrac{1}{2^\lambda}) && (3.34)\\
&\geq \tfrac{1}{2} - \tfrac{1}{2^{\lambda+1}} && (3.35)
\end{aligned}$$

This concludes this proof. □

Now, we define two languages $\mathcal{L}_{cand}, \mathcal{L}_{size}$ as follows. The language $\mathcal{L}_{cand}$ consists of tuples $(\rho, i, b, \{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]})$ containing the the bit i, the value of the bit b of the ρ'th candidate key in the set after filtering the set of secret keys with the tuples $(x_i, y_i, \pi_i, \tau_i)$ for all $i \in [s(\lambda) + \lambda]$. We introduce the notation $\mathrm{rk}_\rho(S)$ for $S \subseteq \{0,1\}^{s(\lambda)} = \mathcal{SK}$ and $\rho \in [|S|]$ to denote the ρ'th element in S in canonical ordering. The language $\mathcal{L}_{size}$ contains tuples $(m, \{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]})$ if m is the size of the set of candidate keys after filtering with the tuples $(x_i, y_i, \pi_i, \tau_i)$ for all $i \in [s(\lambda) + \lambda]$.

$$\begin{aligned}
\mathcal{L}_{cand} := \Bigg\{ &\Big(\rho, i, b, \{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]}\Big) : \\
&\mathrm{rk}_\rho\Big(\textstyle\bigcap_{j\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_j, y_j, \pi_j, \tau_j)\Big)_i = b \Bigg\}
\end{aligned} \tag{3.36}$$

$$\begin{aligned}
\mathcal{L}_{size} := \Bigg\{ &\Big(m, \{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]}\Big) : \\
&\Big|\textstyle\bigcap_{j\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_j, y_j, \pi_j, \tau_j)\Big| \leq m \Bigg\}
\end{aligned} \tag{3.37}$$

We show that $\mathcal{L}_{cand}$ and $\mathcal{L}_{size}$ are in PSPACE and therefore membership can be decided by the PSPACE complete oracle. This will allow the adversary to decide membership using its PSPACE complete oracle.

Claim 3.17 *$\mathcal{L}_{cand} \in$ PSPACE and $\mathcal{L}_{size} \in$ PSPACE.*

Proof (Claim 3.17) We show membership in PSPACE by providing deterministic Turing machines M_{cond} and M_{size} that decide $\mathcal{L}_{cond}$ and $\mathcal{L}_{size}$ respectively such that they require space polynomial in their input. Before we provide the Turing machines M_{cond} and M_{size} we first show that $\mathsf{EvalSim}_{\mathsf{VRF}}$ can be executed in space polynomial in the size of the input.

To run $\mathsf{EvalSim}_{\mathsf{VRF}}$ in space polynomial in the input $(\mathsf{sk}, x, y, \pi, \tau)$ we simulate the evaluation algorithm implemented by the combiner DAG. First compute the topological ordering in polynomial space, then initialize a counter c and simulate the combiner answering the obscure queries with the tuple of assignments corresponding to the c'th element in the canonical on the tuple of well-formed assignments of the obscure queries. That way we only need to store the values of all intermediate results and queries of a single combiner evaluation execution, since this execution runs in polynomial time, every iteration only requires polynomial space.

To show $\mathcal{L}_{cond} \in$ PSPACE consider the following Turing machine M_{cond}. On input $(\rho, i, b, \{x_j, y_j, \pi_j, \tau_j\}_{j\in[s(\lambda)+\lambda]})$ M_{cond} starts by enumerating all secret keys $\mathsf{sk} \in \mathcal{SK}$ in lexicographic order. For every sk iterate over every tuple $(x_j, y_j, \pi_j, \tau_j)$ over $j \in [s(\lambda) + \lambda]$ and run $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_j, y_j, \pi_j, \tau_j)$. If the invocation returns $\perp$ discard the secret key and move onto the next onw. Otherwise after we iterated over all $s(\lambda) + \lambda$ tuples increase a counter c by one until $c = \rho$. If $c = \rho$ and $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_j, y_j, \pi_j, \tau_j) = \top$ then look at the i'th bit b' and accept if and only if $b' = b$.

To show $\mathcal{L}_{size} \in$ PSPACE consider the following Turing machine M_{size}. On input $(m, \{x_j, y_j, \pi_j, \tau_j\}_{j\in[s(\lambda)+\lambda]})$ M_{cond} proceeds in a similar fashion to M_{cond}. M_{soze} starts by enumerating all secret keys $\mathsf{sk} \in \mathcal{SK}$ in lexicographic order, for every sk we iterate over the tuples $(x_j, y_j, \pi_j, \tau_j)$ for $j \in [s(\lambda) + \lambda]$ and run and run $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_j, y_j, \pi_j, \tau_j)$. If the invocation returns $\perp$ discard the secret key and move onto the next one. Otherwise after we iterated over all $s(\lambda) + \lambda$ tuples increase a counter c by one. After all secret keys return c. □

We now describe the adversary $\mathcal{A}$ that wins the pseudorandomness security experiment against the combiner initialized with oracles $\mathcal{O}_{vrf}$ and $\mathcal{O}_{uppc}$ with non-negligible advantage. First, $\mathcal{A}$ first samples $s(\lambda) + \lambda$ evaluation points uniformly at random $x_1, \ldots, x_{s(\lambda)+\lambda}$. Subsequently, $\mathcal{A}$ uses the evaluation oracle in the prf game to evaluate the points to obtain $(y_1, \pi_1), \ldots, (y_{s(\lambda)+\lambda}, \pi_{s(\lambda)+\lambda})$. Using the public key pk^* the adversary will then compute the combiner's verifier traces for all tuples consisting of evaluation point and image-proof pair $(x_1, y_1, \pi_1), \ldots, (x_{s(\lambda)+\lambda}, y_{s(\lambda)+\lambda}, \pi_{s(\lambda)+\lambda})$ to

obtain $\tau_1, \ldots, \tau_{s(\lambda)+\lambda}$. Next, the adversary will use binary search over m to find the size of the candidate key set, k, by querying the PSPACE oracle to decide membership in $\mathcal{L}_{size}$ before sampling a random element $\rho \leftarrow_{\$} [k]$ and recovering the ρ'th candidate sk_{cand} bit-by-bit using the PSAPCE-complete oracle to decide membership in $\mathcal{L}_{cand}$. Finally, the adversary samples a random challenge pre-image $x^* \leftarrow_{\$} \{0,1\}^\lambda \setminus \{x_1, \ldots, x_{s(\lambda)+\lambda}\}$, obtain from the challenger the challenge image y^*, computes the image y' of x^* under the combiner's evaluation algorithm with the secret key candidate sk_{cand} and guesses bit $b = 0$ if and only if $y' = y^*$. We illustrate the adversary in Figure 3.3.

Claim 3.18 *Let $\mathcal{A}$ be the adversary as described above and in Figure 3.3. $\mathcal{A}$ runs in time polynomial in the security parameter λ and wins the pseudorandomness security experiment against any robust black-box DAG combiner Π_{VRF} instantiated with the candidates $\mathcal{O}_{vrf}$ and $\mathcal{O}_{up}$ as above with non-negligible advantage. Specifically*

$$\mathbf{Adv}^{\text{prf}}_{\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}}(\mathcal{A}, \lambda) \geq \left| \frac{1}{|\mathcal{Y}|} - \frac{1}{2} + \frac{1}{2^{\lambda+1}} \right|$$

Proof (Claim 3.18) *Runtime*: The adversary $\mathcal{A}$ can sample $s(\lambda) + \lambda$ evaluation points, query the games evaluation oracle on them to obtain the image-proof pairs and evaluate the combiner's verification algorithm on the point-image-proof tuples to obtain the trace in time $\mathcal{O}(s(\lambda) + \lambda)$. The set of candidate keys is bounded in size by the size of the secret key space, thus by means of binary search $\mathcal{A}$ can obtain the size with at most $s(\lambda)$ queries to the PSPACE complete oracle. With another $s(\lambda)$ queries $\mathcal{A}$ can then recover the ρ'th candidate key bit-by-bit. The final evaluation of the combiner's evaluation algorithm on the candidate secret key and the challenge pre-image can also be done in polynomial time due to the efficiency requirement on the robust black-box DAG combiners. Overall, the adversary runs in time polynomial in λ and n.

Advantage: Since the proof of the advantage bound is lengthy, we defer it to the appendix. We state the result as a claim in Claim 3.19.

Claim 3.19 *Let $\mathcal{A}$ be the adversary as in Figure 3.3. Then*

$$\mathbf{Adv}^{\text{prf}}_{\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}}(\mathcal{A}, \lambda) \geq \left| \frac{1}{|\mathcal{Y}|} - \frac{1}{2} + \frac{1}{2^{\lambda+1}} \right|$$

Using Claim 3.19 concludes this proof. Note, that if $|\mathcal{Y}| \in \mathsf{poly}(\lambda)$ then the scheme would trivially admit efficient adversaries against the pseudorandomness security experiment. Therefore, we can assume that $1/|\mathcal{Y}| \in \mathsf{negl}(\lambda)$ which then results a non-negligible advantage bound for $\mathcal{A}$. □

Using Claim 3.18 we can now finally prove the Theorem 3.5.

Adversary $\mathcal{A}_1^{\mathsf{PSPACE},\mathcal{O}_{eval},\mathcal{O}_{vrf},\mathcal{O}_{uppc}}(\mathsf{pk})$

1 : // Obtain $\{(x_i, y_i, \pi_i, \tau_i)\}_{i\in[s(\lambda)+\lambda]}$:
2 : $\mathcal{X} \leftarrow \varnothing$
3 : **for** $i \in [s(\lambda)+\lambda]$ **do**
4 : $\quad x_i \leftarrow_{\$} \{0,1\}^{\lambda}$
5 : $\quad \mathcal{X} \leftarrow \mathcal{X} \cup \{x_i\}$
6 : $\quad (y_i, \pi_i) \leftarrow \mathcal{O}_{eval}(x_i)$
7 : $\quad \tau_i \leftarrow \mathsf{trace}(\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x_i, y_i, \pi_i))$
8 : // Obtain $|\bigcap_{i\in[s(\lambda)+\lambda]} \mathcal{SK}_{cand}(x_i, y_i, \pi_i, \tau_i)|$ using binary search:
9 : $\ell \leftarrow 0$
10 : $u \leftarrow 2^{s(\lambda)}$
11 : **while** $\ell < u$ **do**
12 : $\quad m \leftarrow \lfloor(\ell+u)/2\rfloor$
13 : $\quad b \leftarrow \mathsf{PSPACE}\big((m, \{x_i, y_i, \pi_i, \tau_i\}_{i\in[s(\lambda)+\lambda]}) \in \mathcal{L}_{size}\big)$
14 : $\quad$ **if** $b = \top$ **then**
15 : $\quad\quad u \leftarrow m$
16 : $\quad$ **else**
17 : $\quad\quad \ell \leftarrow m+1$
18 : $k \leftarrow \ell$
19 : $\rho \leftarrow_{\$} [k]$
20 : // Obtain $\mathsf{sk}_{cand} \in \mathcal{SK}_{cand}$:
21 : $\mathsf{sk}_{cand} \leftarrow \varepsilon$
22 : **for** $i \in [s(\lambda)]$ **do**
23 : $\quad b \leftarrow \mathsf{PSPACE}\big((0, \rho, i, \{x_i, y_i, \pi_i, \tau_i\}_{i\in[s(\lambda)+\lambda]}) \in \mathcal{L}_{cand}\big)$
24 : $\quad$ **if** $b = \top$ **then**
25 : $\quad\quad \mathsf{sk}_{cand} \leftarrow \mathsf{sk}_{cand} \| 0$
26 : $\quad$ **else if** $b = \bot$ **then**
27 : $\quad\quad \mathsf{sk}_{cand} \leftarrow \mathsf{sk}_{cand} \| 1$
28 : $x^* \leftarrow_{\$} \{0,1\}^{\lambda} \setminus \mathcal{X}$
29 : **return** $(\mathsf{sk}_{cand}, x^*)$

Adversary $\mathcal{A}_2^{\mathsf{PSPACE},\mathcal{O}_{eval},\mathcal{O}_{vrf},\mathcal{O}_{uppc}}(\mathsf{pk}, \mathsf{sk}_{cand}, x^*, y^*)$

1 : $y' \leftarrow \Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x^*)$
2 : **return** $y' = y^*$

Figure 3.3: Adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ against $\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}$. Besides the candidate schemes $\mathcal{O}_{vrf}$ and $\mathcal{O}_{uppc}$ the oracle $\mathcal{O}_{eval}$ is the evaluation oracle from the pseudorandoness security experiment and PSPACE is a PSPACE complete oracle. We use $\|$ to denote concatenation.

Proof (Theorem 3.5) Claim 3.18 states that for every robust $(2;1)$ black-box DAG VRF-combiner Π_{VRF} there exists a polynomial time adversary $\mathcal{A}$ with non-negligible advantage against Π_{VRF} initialized with candidates $\mathcal{O}_{vrf}$ and $\mathcal{O}_{up}$. Since $\mathcal{O}_{vrf}$ implements a secure VRF we conclude that there exists no robust $(2;1)$ black-box DAG VRF combiner since there exists no polynomial time adversary $\mathcal{B}$ against $\mathcal{O}_{vrf}$. Since there exists no robust $(2;1)$ black-box DAG VRF-combiner, by Theorem 3.2, there also doesn't exist a robust $(2;1)$ black-box VRF-combiner. □

While robust $(2;1)$ black-box combiners are generally sufficient to construct robust $(n;1)$ black-box combiners, non-existence of a robust $(2;1)$ black-box combiner only allows one to deduce the impossibility of a robust $(n;\lfloor n/2 \rfloor)$ black-box combiner. This allows us to extend the result in Theorem 3.5 to an infinite family of robust black-box VRF-combiners.

Lemma 3.20 (Robust $(n;\lfloor n/2 \rfloor)$ Black-Box VRF Combiner) *There exists no robust $(n;\lfloor n/2 \rfloor)$ black-box combiner in the sense of Definition 2.12 for the VRF primitive defined in Definition 2.9 for all $n \in \mathbb{N}^+$.*

Proof (Lemma 3.20) We show how to construct a robust $(2;1)$ black-box VRF-combiner $\widetilde{\Pi}_{\mathsf{VRF}}$ from a robust $(n;\lfloor n/2 \rfloor)$ black-box VRF-combiner Π_{VRF}. The result holds by contraposition of this fact applied to Theorem 3.5. $\widetilde{\Pi}_{\mathsf{VRF}}$ is simply Π_{VRF} but the first $\lfloor n/2 \rfloor$ candidates are initialized with $\mathcal{O}_1$ and the other $\lceil n/2 \rceil$ candidates are initialized with $\mathcal{O}_2$. If either $\mathcal{O}_1$ or $\mathcal{O}_2$ implements a secure VRF then at least $\lfloor n/2 \rfloor$ candidates used to initialize Π_{VRF} are secure, the initialized combiner is secure. □

Extending this result to an impossibility of robust $(n;n-1)$ turned out to be more difficult than expected. While the proof for Theorem 3.5 doesn't seem to generalize to an impossibility result for robust $(n;n-1)$ robust black-box VRF-combiner we still believe that such constructions can be proven to be impossible. We state the result as a conjecture and defer to the appendix for a discussion on why the approaches don't easily generalize.

Conjecture 3.21 *There exists no robust $(n;n-1)$ black-box combiner in the sense of Definition 2.12 for the VRF primitive as defined in Definition 2.9 for all positive natural numbers $n \in \mathbb{N}^+$.*

Finally we want to note that the adversary constructed in the proof of Theorem 3.5 can predict the image of any arbitrary pre-image $x \in \{0,1\}^\lambda$. This means that the adversary is strong enough win the weak pseudorandomness and unpredictability security experiment with the same advantage as in Claim 3.18. As a corollary of Theorem 3.5, we get the impossibility of robust $(n;\lfloor n/2 \rfloor)$ black-box combiners for the VUF and WVRF primitive.

Corollary 3.22 (Robust $(n; \lfloor n/2 \rfloor)$ Black-Box VUF-combiner) *There exists no robust $(n; \lfloor n/2 \rfloor)$ black-box combiner in the sense of Definition 2.12 for the VUF primitive as defined in Definition 2.11.*

Corollary 3.23 (Robust $(n; \lfloor n/2 \rfloor)$ Black-Box WVRF-combiner) *There exists no robust $(n; \lfloor n/2 \rfloor)$ black-box combiner in the sense of Definition 2.12 for the WVRF primitive as defined in Definition 2.10.*

3.3 Relaxed Robust Black-Box VRF-Combiners

In the previous section we established an impossibility of robust $(n; \lfloor n/2 \rfloor)$ black-box combiners for the VRF, WVRF and VUF primitives and conjectured that the result extends to robust $(n; n-1)$ black-box combiners. A natural follow-up question is whether the impossibility results can be circumvented by either relaxing the requirements or imposing additional assumptions on the candidates. In this section, we show that under the additional assumption that all candidates satisfy unique provability and perfect correctness, there exist simple robust $(n; 1)$ black-box combiners for VRFs.

We argue that the additional assumptions are reasonable in practice. Since perfect correctness and unique provability are information-theoretic properties they don't rely on computational hardness assumptions but can generally be proved rigorously. As a result, for concrete VRF constructions that satisfy those properties, the guarantee holds regardless of advances in cryptography and therefore there is no reason to hedge against them by using combiners.

As already mentioned in Section 1.2, PRFs admit a simple robust $(n; 1)$ black-box combiner which simply XORs the outputs of all PRF candidates. We show that this construction can be extended to VRFs by simply computing the transcript of the evaluation algorithms and providing this as a proof.

Lemma 3.24 (Relaxed Robust $(n; 1)$ Black-Box VRF Combiners) *There exist robust $(n; 1)$ black-box combiners in the sense of Definition 2.12 for the VRF primitive Definition 2.9 if we assume that all candidates are uniquely provable in the sense of Definition 2.5 and perfectly correct in the sense of Definition 2.4.*

We construct a robust $(2; 1)$ black-box VRF-combiner under the assumption that all candidates are perfectly correct and uniquely provable. A robust $(n; 1)$ black-box VRF-combiner can then be constructed by iteratively applying the construction, combining the previous combiner's output with the next candidate oracle until all n candidate schemes are incorporated.

The construction, illustrated in Figure 3.4, closely follows the construction of the robust $(2; 1)$ black-box PRF-combiner in Section 1.2. The key generation algorithm samples a key pair per candidate and combines the secret and public keys to form the combined key pair. The evaluation algorithm evaluates the point x with both candidate schemes to obtain (y_1, π_1) and (y_2, π_2). The image is then $y = y_1 \oplus y_2$ with the proof being $\pi = (y_1, \pi_1, y_2, \pi_2)$. The verification algorithm works by first checking whether both values y_1, y_2 with their respective proofs π_1, π_2 verify under the verification algorithm of the respective candidate. Secondly, it checks whether $y = y_1 \oplus y_2$.

Claim 3.25 *The combiner in Figure 3.4 is perfectly correct (Definition 2.4) and uniquely provable (Definition 2.5) if all candidates are perfectly correct (Definition 2.4) and uniquely provable (Definition 2.5).*

Algorithm $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$

1 : $(\mathsf{sk}_1, \mathsf{pk}_1) \leftarrow_\$ \mathcal{O}_1.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$
2 : $(\mathsf{sk}_2, \mathsf{pk}_2) \leftarrow_\$ \mathcal{O}_2.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$
3 : **return** $((\mathsf{sk}_1, \mathsf{sk}_2), (\mathsf{pk}_1, \mathsf{pk}_2))$

Algorithm $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$

1 : $(\mathsf{sk}_1, \mathsf{sk}_2) \leftarrow \mathsf{sk}$
2 : $(y_1, \pi_1) \leftarrow \mathcal{O}_1.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_1, x)$
3 : $(y_2, \pi_2) \leftarrow \mathcal{O}_2.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_2, x)$
4 : $y \leftarrow y_1 \oplus y_2$
5 : $\pi \leftarrow (y_1, \pi_1, y_2, \pi_2)$
6 : **return** (y, π)

Algorithm $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)$

1 : $(\mathsf{pk}_1, \mathsf{pk}_2) \leftarrow \mathsf{pk}$
2 : $(y_1, \pi_1, y_2, \pi_2) \leftarrow \pi$
3 : $b_1 \leftarrow \mathcal{O}_1.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_1, x, y_1, \pi_1)$
4 : $b_2 \leftarrow \mathcal{O}_2.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_2, x, y_2, \pi_2)$
5 : **if** $y = y_1 \oplus y_2 \wedge b_1 = 1 \wedge b_2 = 1$ **then**
6 : **return** 1
7 : **else**
8 : **return** 0

Figure 3.4: Relaxed robust $(2;1)$ black-box VRF-combiner Π_{VRF} for uniquely provable and perfectly correct candidates $\mathcal{O}_1, \mathcal{O}_2$.

Proof (Claim 3.25) *Correctness:* Since both $\mathcal{O}_1$ and $\mathcal{O}_2$ are assumed to be correct, the image-proof pairs (y_1, π_1) and (y_2, π_2) computed are accepted by their respective verification algorithm. By construction of the combiner $y = y_1 \oplus y_2$. Therefore, an honestly produced image-proof pair is accepted by the combiner's verification algorithm.

Unique Provability: Assume towards a contradiction that there exist image-proof pairs (y_1, π_1) and (y_2, π_2) with $y_1 \neq y_2$, such that both are accepted by the combiner's verification algorithm under a common public key pk. By construction, we know $y_1 = y_{11} \oplus y_{12}$, $y_2 = y_{21} \oplus y_{22}$, $\pi_1 = (y_{11}, \pi_{11}, y_{12}, \pi_{12})$ and $\pi_2 = (y_{21}, \pi_{21}, y_{22}, \pi_{22})$ (otherwise the verification algorithm would reject). Since $y_1 \neq y_2$ we have that $(y_{11}, y_{12}) \neq (y_{21}, y_{22})$. W.l.o.g. let $y_{11} \neq y_{21}$. Since both (y_1, π_1) and (y_2, π_2) are accepted by the verification algorithm, by construction of the combiner, there must exist π_{11} and π_{21} such that the verification algorithm of $\mathcal{O}_1$ both accepts (y_{11}, π_{11}) and (y_{21}, π_{21}). This is a contradiction with the assumed unique provability of $\mathcal{O}_1$. □

Claim 3.26 *Consider a combiner Π_{VRF} as defined in Figure 3.4 instantiated with candidates $\mathcal{O}_1, \mathcal{O}_2$. If at least one candidate is (t, ε)-pseudorandom (Definition 2.6),*

then the combiner is $(\mathfrak{t}, \varepsilon)$*-pseudorandom. In particular, if at least one candidate is pseudorandom, then the combiner is pseudorandom.*

W.l.o.g., assume that $\mathcal{O}_1$ is $(\mathfrak{t}, \varepsilon)$-pseudorandom for $\varepsilon \in \mathsf{poly}(\lambda)$ and $\mathfrak{t} \in \mathsf{poly}(\lambda)$. Let $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ be an adversary against the combiner $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ running in time at most $\mathfrak{t}$. We construct an adversary $\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2)$ that uses $\mathcal{A}$ to break the pseudorandomness of $\mathcal{O}_1$ with the same advantage. The adversary $\mathcal{B}$ is shown in Figure 3.5.

$\mathcal{B}$ samples its own key-pair $(\mathsf{sk}, \mathsf{pk})$ for the candidate $\mathcal{O}_2$ and uses $\mathcal{O}_{\text{Eval}}$ to simulate the combiner $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ in the pseudorandomness security game of $\mathcal{A}$. Whenever $\mathcal{A}$ queries an input x to its evaluation oracle, $\mathcal{B}$ queries x to its own evaluation oracle to obtain (y, π) then computes the output

Assume, towards a contradiction, that there exists an adversary $\mathcal{A}$ running in time $\mathfrak{t}$ that breaks the pseudorandomness of the combiner $\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}$ with advantage ε, we show that we can construct an adversary $\mathcal{R}_i$ running in time t_i that wins the prf security experiment of the i'th candidate $\mathcal{O}_i$ with advantage ε_i for $i \in [2]$. The reduction $\mathcal{R}_i$ samples its own secret key for candidate $\mathcal{O}_{2-i}$ and uses $\mathcal{O}_{eval}$ for candidate $\mathcal{O}_i$ simulate the combiner in the prf security experiment for the adversary $\mathcal{A}$. $\mathcal{A}$ then makes a prediction for the hidden bit. If the hidden bit of the challenger is 0 then $\mathcal{A}$ is simulated perfectly in the sense that the queries are answered exactly as they would have been answered in the combiners prf security experiment. In that case $\mathcal{R}_i$ wins the experiment if and only if $\mathcal{A}$ wins. If $b = 1$ the output of $\mathcal{O}_{eval}$ is a uniformly random element in the image codomain and therefore the queries made by $\mathcal{A}$ are also answered with a uniformly random element in the codomain. Also in this case $\mathcal{A}$ is simulated perfectly. and uses adversary $\mathcal{A}$ to break that one candidate. We show the reduction $\mathcal{R}_i$ in Figure 3.5.

Reduction $\mathcal{R}_i^{\mathcal{O}_{i_{eval}}}(\mathsf{pk}_i)$

$b \leftarrow_{\$} \{0,1\}$
$(\mathsf{sk}_{2-i}, \mathsf{pk}_{2-i}) \leftarrow_{\$} \mathcal{O}_{eval,2-i}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$
$\mathsf{sk} \leftarrow (\mathsf{sk}_1, \mathsf{sk}_2)$
$\mathsf{pk} \leftarrow (\mathsf{pk}_1, \mathsf{pk}_2)$
$S \leftarrow \varnothing$
$(\sigma, x^*) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{comb_{eval}}}(\mathsf{pk}_1, \mathsf{pk}_2)$
$y_1 \leftarrow_{\$} \mathcal{Y}$
$b' \leftarrow_{\$} \mathcal{A}(1^\lambda, x^*, \sigma, y_b)$
return $b = b' \wedge x^* \notin S$

Oracle $\mathcal{O}_{comb_{eval}}(x)$

$S \leftarrow S \cup \{x\}$
$(y_i, \pi_i) \leftarrow \mathcal{O}_{i_{eval}}(x)$
$(y_{2-i}, \pi_{2-i}) \leftarrow \mathcal{O}_{2-i}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_{2-i}, x)$
$y \leftarrow y_1 \oplus y_2$
$\pi \leftarrow (y_1, \pi_1, y_2, \pi_2)$
return (y, π)

Figure 3.5: Reduction.

The reduction $\mathcal{R}_i$ makes as many queries to its evaluation oracle $\mathcal{O}_{i_{eval}}$ as $\mathcal{A}$ does to its evaluation oracle $\mathcal{O}_{comb_{eval}}$ and makes additionally as many queries to the oracle $\mathcal{O}_{2-i}$. Besides the oracle queries every computation performed by $\mathcal{R}_i$ is efficiently computable. Since we assume $\mathcal{A}$ to be efficient (ppt.) the reduction $\mathcal{R}_i$ must also be efficient (ppt.).

The advantage of the reduction $\mathcal{R}_i$ in the prf security experiment against candidate $\mathcal{O}_i$ can be bounded from below as follows.

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathsf{i}}(\lambda, \mathcal{R}_i) = \mathbf{Adv}^{\mathrm{prf}}_{\mathsf{comb}}(\lambda, \mathcal{A}) \tag{3.38}$$

Proof (Lemma 3.24) Follows directly from Claim 3.25 and Claim 3.26 and□

Chapter 4

Conclusion

We introduced robust black-box DAG combiners that make the structure of robust black-box combiners more explicit while capturing the same set of constructions. Subsequently, we answered the question whether there exists a robust $(2;1)$ black-box VRF-combiner negatively. The result implies an impossibility result on the family of robust $(n;\lfloor n/2 \rfloor)$ black-box VRF-combiners, placing a lower bound on the number of secure instances needed to combine VRFs in a robust way.

On the other hand, we show that if we strengthen the assumptions on the candidate schemes, specifically by requiring all of them to be uniquely provable and perfectly correct, then VRFs admit simple combiners similar to the known PRF combiners.

Extensions. While we strongly believe robust $(n;n-1)$ black-box VRF-combiners to be impossible, extending the proof for the impossibility of robust $(2;1)$ black-box VRF-combiners to the more general case seems difficult. We pose this question as a conjecture.

Additionally, it is natural to think about weaker relaxations of robust black-box VRF-combiners that circumvent the impossibility result, beyond the one we presented. For instance, it is interesting to study the possibility of combining VRFs if the combiner only needs to satisfy statistical correctness.

Finally, even for the strong relaxation of VRF-combiners that admit simple XOR combiners, it is interesting to study relaxed robust black-box combiners for more advanced notions of VRFs such as constrainable VRFs [Fuc14], key-homomorphic VRFs [Mal24], aggregable VRFs [Mal24] and unbiasable VRFs [GS24].

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Appendix

Deferred Proofs

Proof of Claim 3.9

Claim .1 *Let* $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \widetilde{\mathcal{O}}_{uppc}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$ *and* $\widetilde{\mathcal{SK}} \subseteq \mathcal{SK}$ *and* $\widetilde{\mathcal{PK}} \subseteq \mathcal{PK}$ *be arbitrary subsets. Then the following holds:*

$$\mathsf{sk} \in cl_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) \iff \mathsf{pk} \in cl_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$$

Proof (Claim 3.9) We show $\mathsf{sk} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}}) \implies \mathsf{pk} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$:

By definition of the closure, we have:

$$\mathsf{sk} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}) \vee \mathsf{sk} \in \bigcup_{\widetilde{\mathsf{pk}} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})$$

We distinguish two cases: First, assume $\mathsf{sk} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})$, then:

$$\mathsf{pk} \in R_{\mathcal{PK}}(\mathsf{sk})$$

Combining this with $\mathsf{sk} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})$, we get:

$$\mathsf{pk} \in \bigcup_{\widetilde{\mathsf{sk}} \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})} R_{\mathcal{PK}}(\widetilde{\mathsf{sk}})$$

And hence $\mathsf{pk} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$.

On the other hand, if $\mathsf{sk} \notin \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})$, then:

$$\mathsf{sk} \in \bigcup_{\widetilde{\mathsf{pk}} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})$$

In particular, we have $\mathsf{sk} \in R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})$ for some $\widetilde{\mathsf{pk}}$. By definition of $\sim_{\mathcal{PK}}$, we have $\widetilde{\mathsf{pk}} \sim_{\mathcal{PK}} \mathsf{pk}$ since sk is a secret key which can be sampled together with both pk and $\widetilde{\mathsf{pk}}$. Therefore, $\widetilde{\mathsf{pk}} \sim^*_{\mathcal{PK}} \mathsf{pk}$ which means that $\mathsf{pk} \in [\widetilde{\mathsf{pk}}]_{\sim^*_{\mathcal{PK}}}$ for some $\widetilde{\mathsf{pk}} \in \widetilde{\mathcal{PK}}$. In the end we obtain $\mathsf{pk} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$. The proof for the converse works analogously. □

Proof of Claim 3.11

Claim .2 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$2 \cdot \left| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \right] - \frac{1}{2} \right| = \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right|$$

The proof is by equivalence transformations.

$$\begin{aligned}
\mathbf{Adv}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) &= 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \right] - \frac{1}{2} \right| \\
&= 2 \cdot \Big| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 0 \right] \Pr[b = 0] \\
&\quad + \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 1 \right] \Pr[b = 1] - \frac{1}{2} \Big| \\
&= 2 \cdot \Big| \frac{1}{2} \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 0 \right] \\
&\quad + \frac{1}{2} \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 1 \right] - \frac{1}{2} \Big| \\
&= \Big| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 0 \right] \\
&\quad + \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 1 \right] - 1 \Big| \\
&= \Big| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 0 \right] \\
&\quad - \left(1 - \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 1 \right] \right) \Big| \\
&= \Big| \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 0 \right] \\
&\quad - \left(1 - \Pr\left[\mathbf{Exp}^{\text{prf}}_{\widetilde{\mathcal{O}}_{uppc}}(\lambda, \mathcal{A}) = 1 \mid b = 1 \right] \right) \Big| \\
&= \left| \Pr\left[H_2^{\lambda,\mathcal{A}} = 1 \right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1 \right] \right|
\end{aligned}$$

The last step holds by definition of $H_2^{\lambda,A} = 1$ and $H_3^{\lambda,\mathcal{A}}$.

Proof of Claim 3.12

Claim .3 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$\left|\Pr\left[H_2^{\lambda,\mathcal{A}} = 1\right] - \Pr\left[H_1^{\lambda,\mathcal{A}} = 1\right]\right| \leq \varepsilon_{vrf}^{prf}$$

We consider the advantage of an arbitrary adversary $\mathcal{A}$ against $\mathcal{O}_{vrf}$ in the pseudorandomness game. We follow derivations analogously to Claim 3.12.

$$\begin{aligned}
\mathbf{Adv}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A}) &= 2\cdot\left|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\right]-\frac{1}{2}\right| \\
&= 2\cdot\Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=0\right]\Pr[b=0] \\
&\quad+\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=1\right]\Pr[b=1]-\frac{1}{2}\Big| \\
&= 2\cdot\Big|\frac{1}{2}\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad+\frac{1}{2}\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=1\right]-\frac{1}{2}\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad+\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=1\right]-1\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad-\left(1-\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=1\right]\right)\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad-\left(1-\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A})=1\mid b=1\right]\right)\Big| \\
&= \left|\Pr\left[H_2^{\lambda,\mathcal{A}}=1\right]-\Pr\left[H_1^{\lambda,\mathcal{A}}=1\right]\right|
\end{aligned}$$

The last step holds by definition of $H_2^{\lambda,A} = 1$ and $H_1^{\lambda,\mathcal{A}}$.

The bound follows from $\mathbf{Adv}^{\mathrm{prf}}_{\mathcal{O}_{vrf}}(\lambda,\mathcal{A}) \leq \varepsilon_{vrf}^{prf}$.

Proof of Claim 3.13

Claim .4 *Let the games H_1, H_2, H_3 be defined as above, the candidates $\mathcal{O}_{vrf}$ and $\widetilde{\mathcal{O}}_{uppc}$ be defined as above. Then for all adversaries $\mathcal{A}$ we have:*

$$\left|\Pr\left[H_1^{\lambda,\mathcal{A}} = 1\right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1\right]\right| \leq \mu(\lambda)$$

Where $\mu \in \mathsf{negl}(\lambda)$.

For this result we rely on the difference lemma which we state as follows.

Lemma .5 (Difference Lemma) *Let $\mathcal{W}_0, \mathcal{W}_1, \mathcal{E}$ be events with*

$$\Pr[\mathcal{W}_0 \wedge \neg\mathcal{E}] = \Pr[\mathcal{W}_1 \wedge \neg\mathcal{E}] \tag{.1}$$

Then we can bound the difference in probability of $\mathcal{W}_0$ and $\mathcal{W}_1$ with

$$|\Pr[\mathcal{W}_0] - \Pr[\mathcal{W}_1]| \leq \Pr[\mathcal{E}]$$

Proof (Lemma .5) Let $\mathcal{W}_0, \mathcal{W}_1, \mathcal{E}$ be events satisfying (.1). Then:

$$\begin{aligned}
|\Pr[\mathcal{W}_0] - \Pr[\mathcal{W}_1]| =& \big|\Pr[\mathcal{W}_0 \wedge \mathcal{E}] + \Pr[\mathcal{W}_0 \wedge \neg\mathcal{E}] \\
&\quad - \Pr[\mathcal{W}_1 \wedge \mathcal{E}] - \Pr[\mathcal{W}_1 \wedge \neg\mathcal{E}]\big| \\
=& |\Pr[\mathcal{W}_0 \wedge \mathcal{E}] - \Pr[\mathcal{W}_1 \wedge \mathcal{E}]| \\
=& |\Pr[\mathcal{W}_0 \mid \mathcal{E}]\Pr[\mathcal{E}] - \Pr[\mathcal{W}_1 \mid \mathcal{E}]\Pr[\mathcal{E}]| \\
=& |\Pr[\mathcal{W}_0 \mid \mathcal{E}] - \Pr[\mathcal{W}_1 \mid \mathcal{E}]|\Pr[\mathcal{E}] \\
\leq& \Pr[\mathcal{E}]
\end{aligned}$$

□

Note that if the secret key sampled by the challenger in the games H_1 and H_3 satisfies $\mathsf{sk}^* \notin \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$, then the games are identical. This is because $\widetilde{\mathcal{O}}_{uppc}$ evaluates to the same images as $\mathcal{O}_{vrf}$ on all $\mathsf{sk}^* \notin \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$. We can now use Lemma .5 to bound the difference in probabilities:

$$\left|\Pr\left[H_1^{\lambda,\mathcal{A}} = 1\right] - \Pr\left[H_3^{\lambda,\mathcal{A}} = 1\right]\right| \leq \Pr\left[\mathsf{sk}^* \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})\right] \tag{.2}$$

Applying the definition of $\mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}}, \widetilde{\mathcal{PK}})$ and subsequently apply the union

bound we obtain after further simplifications:

$$\left|\Pr\left[H_1^{\lambda,\mathcal{A}}=1\right]-\Pr\left[H_3^{\lambda,\mathcal{A}}=1\right]\right| \leq \Pr\left[\mathsf{sk}^* \in \mathsf{cl}_{\mathcal{SK}}(\widetilde{\mathcal{SK}})\right] \tag{.3}$$

$$+\Pr\left[\bigcup_{\widetilde{\mathsf{pk}}\in\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})\right] \tag{.4}$$

$$\leq \Pr\left[\mathsf{sk}^* \bigcup_{\widetilde{\mathsf{sk}}\in\widetilde{\mathcal{SK}}} [\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}\right] \tag{.5}$$

$$+\Pr\left[\bigcup_{\widetilde{\mathsf{pk}}\in\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})\right] \tag{.6}$$

$$\leq \sum_{\widetilde{\mathsf{sk}}\in\widetilde{\mathcal{SK}}} \Pr\left[\mathsf{sk}^* \in [\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}\right] \tag{.7}$$

$$+\sum_{\widetilde{\mathsf{pk}}\in\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} \Pr\left[R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})\right] \tag{.8}$$

$$\leq \sum_{\widetilde{\mathsf{sk}}\in\widetilde{\mathcal{SK}}} \Pr\left[\mathsf{sk}^* \in [\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}\right] \tag{.9}$$

$$+\sum_{\widetilde{\mathsf{pk}}\in\mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathcal{PK}})} \Pr\left[R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})\right] \tag{.10}$$

$$\tag{.11}$$

We bound the two terms separately. For each term, we show that if the corresponding probability were non-negligible, we could construct an adversary that breaks pseudorandomness of $\mathcal{O}_{vrf}$.

Consider the first sum. For each $\widetilde{\mathsf{sk}} \in \widetilde{\mathcal{SK}}$, the equivalence class $[\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}$ contains secret keys that are related via the transitive closure of $\sim_{\mathcal{SK}}$. By definition of $\sim_{\mathcal{SK}}$, two secret keys are related if they can be sampled with the same public key. Since $\mathcal{O}_{vrf}$ is uniquely provable, for any public key pk there is at most one valid image per input x. Therefore, all secret keys in $[\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}$ that could be sampled with the same public key must produce the same image on any input. This means that the number of distinct images produced by secret keys in $[\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}$ is bounded by the number of equivalence classes under the relation induced by sharing a public key.

Suppose $\Pr[\mathsf{sk}^* \in [\widetilde{\mathsf{sk}}]_{\sim^*_{\mathcal{SK}}}] \geq \epsilon(\lambda)$ for some non-negligible ϵ. We construct an adversary $\mathcal{B}$ against the pseudorandomness of $\mathcal{O}_{vrf}$ as follows: $\mathcal{B}$ receives a public key pk^* and a challenge (x^*, y^*) where y^* is either $\mathcal{O}_{vrf}.\mathsf{Eval}^y_{\mathsf{VRF}}(\mathsf{sk}^*, x^*)$ or uniformly random. Since $|\widetilde{\mathcal{SK}}| \in \mathsf{poly}(\lambda)$, the adversary $\mathcal{B}$ can simply guess the secret key $\mathsf{sk}_{cand} \in \widetilde{\mathcal{SK}}$ and verify whether

$\mathcal{O}_{vrf}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x^*) = y^*$. With non-negligible probability $\mathcal{B}$ samples a secret key that is in the same equivalence class as sk^*, the game's secret key. In that case, the adversary always wins. Hence, $\mathcal{B}$ has non-negligible advantage in the pseudorandomness security experiment against $\mathcal{O}_{up}$, contradicting the assumed pseudorandomness of $\mathcal{O}_{vrf}$.

The argument for the second sum proceeds analogously. For every $\widetilde{\mathsf{pk}} \in \mathsf{cl}_{\mathcal{PK}}(\widetilde{\mathsf{pk}})$ the set $R_{\mathcal{SK}}(\widetilde{\mathsf{pk}})$ contains all secret keys that can be sampled together with $\widetilde{\mathsf{pk}}$. By unique provability and perfect correctness of $\mathcal{O}_{vrf}$, all such secret keys must produce the same image on any input x. If $\Pr[\mathsf{sk}^* \in R_{\mathcal{PK}}(\widetilde{\mathsf{pk}})] \geq \varepsilon(\lambda)$ for a non-negligible function ε, we can again construct an adversary that wins the pesudorandomness security experiment against $\mathcal{O}_{vrf}$, resulting in a contradiction. Since the sum of two negligible functions is negligible the claim is established.

Proof of Claim 3.19

Claim .6 *Let $\mathcal{A}$ be the adversary as in Figure 3.3. Then*

$$\mathbf{Adv}^{\text{prf}}_{\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}}(\mathcal{A},\lambda) \geq \left|\frac{1}{|\mathcal{Y}|} - \frac{1}{2} + \frac{1}{2^{\lambda+1}}\right|$$

We define the following games:

We define games H_1, H_2 as follows.

- H_1 is the simple pseudorandomness experiment for the hidden bit $b = 0$ against the combiner $\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}$ instantiated with $\mathcal{O}_{vrf}$ and $\mathcal{O}_{up}$, i.e., the adversary is playing against the prf security game where he is given a challenge image y^* computed from the challenge pre-image x^* using $\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*, x^*)$ where sk^* is the hidden secret key sampled by the challenger.
- H_2 is the simple pseudorandomness experiment for the hidden bit $b = 1$ against the oracle $\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}$ instantiated with $\mathcal{O}_{vrf}$ and $\mathcal{O}_{up}$, i.e., the adversary is playing the prf security game where he is given a challenge image y^* computed from the challenge pre-image x^* using $\Pi^{\mathcal{O}_{vrf},\mathcal{O}_{up}}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*, x^*)$ where sk^* is the hidden secret key sampled by the challenger.

$$
\begin{aligned}
\mathbf{Adv}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A}) &= 2\cdot\left|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\right]-\frac{1}{2}\right| \\
&= 2\cdot\Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=0\right]\Pr[b=0] \\
&\quad+\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=1\right]\Pr[b=1]-\frac{1}{2}\Big| \\
&= 2\cdot\Big|\frac{1}{2}\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad+\frac{1}{2}\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=1\right]-\frac{1}{2}\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad+\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=1\right]-1\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad-\left(1-\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=1\right]\right)\Big| \\
&= \Big|\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=0\right] \\
&\quad-\left(1-\Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{up}}}(\lambda,\mathcal{A})=1\mid b=1\right]\right)\Big| \\
&= \Big|\Pr\left[H_1^{\lambda,\mathcal{A}}=1\right]-\Pr\left[H_2^{\lambda,\mathcal{A}}=1\right]\Big| \\
&\geq \left|\frac{1}{2}+\frac{1}{2^{\lambda+1}}-\frac{1}{|\mathcal{Y}|}\right|
\end{aligned}
$$

To justify the last step: During its execution, $\mathcal{A}$ will correctly compute the size of the candidate key set, then obtain a uniformly random candidate secret key. We invoke Claim 3.16 which yields that with probability at least $\frac{1}{2}-\frac{1}{2^{\lambda+1}}$ the candidate secret key will evaluate to the same image on the challenge pre-image x^* as the game's internal secret key would. In the game H_1, the adversary plays against the pseudorandomness game of the combiner with the hidden bit $b=0$. In that case, $\mathcal{A}$ wins if and only if the candidate key and the game's secret key evaluate x^* to the same image y^*. Using Claim 3.16, this happens with probability at least $\frac{1}{2}-\frac{1}{2^{\lambda+1}}$. If the hidden bit is $b=1$, then the adversary will guess bit $b'=1$, except, if the uniformly random image $y^* \leftarrow_{\$} \mathcal{Y}$ is (by chance) equal to the image that the candidate key produces on x^*. This happens with probability $1/|\mathcal{Y}|$.

Conjecture

We briefly remark on the difficulty of extending the proof of Theorem 3.5 to the more general case stated in the conjecture. The direct approach fails due to a subtlety: While we show that the obscure queries are irrelevant on a per-candidate basis, that does not imply the obscure queries are jointly irrelevant over multiple candidates.

In the $(2;1)$ case, we can guarantee that the algorithm $\mathsf{EvalSim}_{\mathsf{VRF}}$ returns $\top$ if the candidate key sk_{cand} is identical to the game's secret key sk^*, precisely because the obscure queries to a single candidate are irrelevant. For $n > 2$ candidates this might, however, not be the case anymore.

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[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).